
Evaluating the Relevance of Uncertainty Estimators for LLM Hallucination

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Abstract

Large language models (LLMs) are prone to hallucinations, i.e., statements unsupported by the input or training data, hindering reliable deployment. In parallel, numerous uncertainty estimation (UE) methods have been proposed to quantify model confidence, and are often implicitly treated as proxies for model failure. However, the relationship between uncertainty and hallucinations remains insufficiently characterized. We present a systematic empirical study of the association between uncertainty estimators and hallucinations in LLMs. Rather than assuming this association, we evaluate directly when and to what extent it holds. We consider a diverse set of uncertainty metrics, including information-theoretic, sampling-based, and reflexive, and examine their behavior across hallucination settings. Our experiments cover both intrinsic hallucinations (violations of input faithfulness) and extrinsic hallucinations (unsupported claims relative to training data), using four complementary benchmarks, including RAGTruth and HalluLens. We find that the association is highly variable and often weak, depending on the hallucination type and the LLM under evaluation. These results challenge the use of uncertainty as a direct signal of hallucination and clarify when it provides actionable information.

1 Introduction

Large language models (LLMs) have demonstrated impressive performance across a wide range of natural language generation tasks, including question answering, summarization, and long-form reasoning [60]. However, despite their fluency and apparent coherence, LLMs often generate hallucinations, i.e., statements that are not supported by the evidence the model is supposed to rely on [4, 16]. Such unsupported generations pose a major obstacle to the safe and reliable deployment of LLMs, particularly in settings where errors can have serious consequences, such as in healthcare, law, and finance [5, 41].

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Recent work has argued that hallucination should be distinguished from factuality, which refers to correctness relative to real-world truth, to better tackle each problem [2, 4]. They further distinguish intrinsic from extrinsic hallucinations, depending on the source of evidence used to assess groundedness. Intrinsic hallucinations occur when model responses are not supported by the information explicitly provided at inference time, whereas extrinsic hallucinations arise when responses are not supported by the knowledge the model is expected to have encountered in its pretraining data [4]. This distinction matters because the mechanisms underlying these two types, and therefore the signals available for detecting them, may differ substantially.

At the same time, substantial effort has been devoted to developing uncertainty estimation (UE) methods for language models, which aim to quantify a model’s confidence in its responses [26, 47]. This includes information-theoretic measures based on token probabilities or likelihoods [18, 34], sampling and ensemble-based approaches that measure variability across generations [27, 31], and reflexive approaches that rely on model self-evaluation [25, 44]. In practice, such signals are often used for abstention or selective prediction and routing uncertain cases to additional verification [1, 55], model self-refinement [56], in-context learning [61], or adaptive retrieval [35].

However, uncertainty estimates are not designed specifically to detect hallucinations, and existing evaluations rarely test them against hallucination-specific targets. Most prior work assesses uncertainty with respect to broader notions of error, correctness, robustness, or calibration rather than with respect to explicitly defined hallucination criteria [3, 47, 51]. Moreover, many uncertainty scores used in practice are heuristic proxies without a rigorous probabilistic foundation.

In this paper, we evaluate whether uncertainty estimators can support hallucination detection when hallucination is defined as inconsistency with expected evidence. We compare 46 estimators across 4 datasets and 3 open-weight instruction-tuned models: Mistral-7B-Instruct, Llama-2-7B-Chat, and Llama-2-13B-Chat. We will often refer to these datasets as *tasks*, as they cover different hallucination and task type: context faithfulness in retrieval-augmented generation (intrinsic) and extrinsic settings, including short-form question answering, long-form generation, and abstention from nonexistent entities. Our main contributions and findings are:

- **A hallucination-specific benchmark for uncertainty.** We evaluate uncertainty estimators against explicit hallucination targets, distinguishing intrinsic context-faithfulness from extrinsic pretraining-data groundedness.
- **The relationship between uncertainty and hallucination depends on the task.** Uncertainty estimators can help separate hallucinated from grounded responses, but their discriminative power varies substantially across datasets. In short-form question answering, most estimators provide strong discrimination. In long-form generation, AUROC values are more uniformly moderate, with smaller differences between estimators. For context-faithfulness and abstention, the discrimination is also moderate, but it is concentrated in fewer estimators and varies more across models. Overall, these results show that the relationship between uncertainty and hallucination depends on the dataset, which encodes not only the hallucination type, but also the task type, response length, evidence source, and expected model behavior. This argues against treating uncertainty scores as general-purpose hallucination detectors.
- **No uncertainty estimator is consistently best across datasets.** The estimators that perform best on one dataset do not reliably remain best on another. Rankings are more stable across models within the same dataset than across datasets for the same model, indicating that dataset variation affects both estimator choice and overall discriminative power. This argues against selecting a hallucination detector from a single aggregate ranking and instead supports dataset-specific validation of uncertainty estimators.
- **Competitive estimators are partially correlated.** Although no uniformly best estimator emerges, the estimators that perform well tend to produce correlated response rankings across datasets and models. As a practical implication, selecting a few representatives from distinct groups of correlated estimators may be preferable to relying on a single estimator or using all available scores.

2 Background on uncertainty estimators

Notation. We consider an autoregressive language model $p_\theta(y \mid x)$, where x denotes the input (query and optional context), $y = (y_1, \dots, y_T)$ is a generated response sequence, and θ parametrizes the model. At each step t , the model produces a token-level probability distribution $p_\theta(\cdot \mid x, y_{<t})$, so that $p_\theta(y \mid x) = \prod_{t=1}^T p_\theta(y_t \mid x, y_{<t})$ for an response sequence of length T , with $y_{<t} = (y_1, \dots, y_{t-1})$.

2.1 Uncertainty definition and scope

There is no universally accepted formal definition of uncertainty in language generation [3, 59]. In this work, uncertainty is treated as a query-level property of the predictive distribution $p_\theta(\cdot \mid x)$. A generic query-level target is denoted $U^*(x) = \Phi(p_\theta(\cdot \mid x))$, where Φ is a property of the predictive distribution [8]. Different estimators may approximate different aspects of this target and need not share the same probabilistic foundation.

Terminology in the literature is not fully consistent: some works use *uncertainty* and *confidence* interchangeably [39, 59], while others reserve *confidence* for response-level uncertainty y [31]. Here, *uncertainty* refers by default to query-level uncertainty. Scores of the form $U(x)$ target this quantity directly, whereas response-conditioned scores $U(x, y)$ are treated as single-response proxies rather than as a separate response-level notion.

2.2 Uncertainty estimators studied

In this work, LLM uncertainty estimators are first organized by access regime, separating white-box [WB] from black-box [BB] methods. Within these settings, the families correspond to groups of estimators with similar input requirements for uncertainty computation. The implementation follows LM-POLYGRAPH [15, 47]; full estimator definitions are given in Section B.

[WB] Information/logit-based methods. Logit- or information-based estimators derive uncertainty directly from the token-level predictive distributions $p_\theta(\cdot \mid x, y_{<t})$ produced during autoregressive generation. They differ in how they aggregate token-level signals over a generated sequence. The simplest methods rely only on the probability of the generated token, yielding sequence-level scores such as negative log-likelihood and related confidence scores based on sequence probability, such as Maximum Sequence Probability (MSP), $1 - p_\theta(y \mid x)$ [18, 34]. Because MSP only uses the probability assigned to the observed sequence, entropy-based methods instead exploit the full token distribution, for example through mean token entropy [18] or similarity-to-uniform measures such as Rényi or Fisher–Rao divergences [11]. Pointwise mutual information methods compare conditional and unconditional token probabilities to quantify the extent to which the query shapes the generated response [46]. More recent variants incorporate semantic information into token probabilities. TokenSAR [13] upweights improbable tokens that are semantically important, while Claim-Conditioned Probability (CCP) focuses uncertainty estimation on token alternatives that preserve or alter the meaning of the current claim [14]. These methods are efficient because they only require a single forward pass to obtain logits, and provide the most direct readout of the model’s predictive distribution.

[WB] Sample-based methods. White-box sample-based methods extend single-pass estimators by repeatedly sampling responses for the same query. Some use these samples to approximate sequence-level uncertainty quantities derived from the model’s predictive distribution, for example through Monte Carlo estimates of entropy or related likelihood-based functionals [34]. However, such estimates do not distinguish semantic variation from surface-form variation across samples. Semantic Entropy, for instance, addresses this limitation by clustering sampled responses into semantic equivalence classes and computing entropy over these discrete classes [27], whereas Semantic Density [39] relies on fine-grained, continuous pairwise semantic similarities instead of a binary semantic-equivalence partition. SentenceSAR [13] further replaces hard semantic clustering with soft semantic weighting, reweighting token-level negative log-likelihood contributions using semantic similarity across sampled responses, so that improbable and semantically isolated responses contribute more strongly to the uncertainty score. Cocoa [48] estimators instead multiply a token logit-based uncertainty score by a semantic-variability score computed from the sampled responses. These methods capture predictive instability beyond a single forward pass but are more computationally expensive and may still fail when the model consistently reproduces the same hallucination.

[WB] Internal-state methods. Internal-state methods exploit signals encoded in the model’s latent computations, using attention patterns or hidden representations as indicators of predictive unreliability. Some measure local processing signals, like AttentionScore that quantify the extent to which a token attends to itself across heads and layers, [43], while others model sequential propagation effects, as in RAUQ [50], or reweight token probabilities using attention-derived saliency [32]. The common premise is that uncertainty may already be reflected in the geometry or dynamics of the forward pass before it becomes visible in the final response distribution.

[WB] Training-based methods. Density- or training-based estimators quantify uncertainty through similarity to the training distribution. They approximate the distribution of training representations, typically using Gaussian densities, and estimate how far an input or representation lies from high-density training regions, often through Mahalanobis distance [29, 49]. These methods are especially relevant for out-of-distribution (OOD) detection and are computationally efficient at inference time, but require access to training data.

[WB] Reflexive methods. Reflexive methods estimate uncertainty by querying the model about its own confidence. Rather than deriving uncertainty from predictive distributions or internal representations, they rely on the model’s explicit self-assessment of the reliability of its answers. One representative example is $P(\text{True})$ prompting, which uses the probability assigned to the token *true* after a follow-up prompt such as “Are you sure?” [25]. These methods are flexible and straightforward to implement, but their reliability depends on the model’s ability to express uncertainty accurately and in a calibrated manner, as well as on the prompt formulation.

[BB] Black-box estimators. Finally, in many practical settings, only generated text is accessible. Black-box methods therefore estimate uncertainty from responses alone, typically by sampling multiple responses, constructing a pairwise lexical or semantic similarity matrix between them, and summarizing its structure into an uncertainty score [31, 36]. While broadly deployable, they are often computationally expensive.

3 Benchmark setup

3.1 Datasets

The evaluation suite spans two types of hallucinations. Intrinsic hallucination is evaluated with RAGTruth [37], where responses are judged against evidence provided at inference time. Extrinsic hallucination is evaluated with three HalluLens datasets [4]: PreciseWikiQA, LongWiki, and NonExistentRefusal, where responses are judged against knowledge expected to be available from pretraining data. Each dataset (task) is associated with a response-level quality metric $Q(x, y)$, derived from its (specific) definition of hallucination. Q serves as the target variable against which uncertainty estimators are assessed: we test whether higher uncertainty is associated with lower response quality.

Intrinsic hallucination: RAGTruth. RAGTruth is a human-annotated dataset for hallucination detection in retrieval-augmented generation [37], spanning summarization, passage-grounded question answering, and data-to-text generation from structured business data. Each instance contains a query, retrieved context, model response, and span-level annotations that mark content that is unsupported by or conflicts with the provided context. We use the processed Hugging Face release², which standardizes metadata and annotations. Under this context-faithfulness definition, information not supported by the retrieved context is treated as hallucinated, even if it is factually correct according to external knowledge. We derive a response-level target from the span annotations: $h_j = 1$ if response y_j contains at least one hallucinated span, and $h_j = 0$ otherwise, with quality target $Q(x_j, y_j) = 1 - h_j$.

Extrinsic hallucination: HalluLens. The three HalluLens datasets evaluate complementary settings of extrinsic hallucination, defined as inconsistency with information expected to be available in the model’s pretraining data [4]. Rather than measuring only generic answer correctness, HalluLens probes deviations from pretraining knowledge: Wikipedia serves as a proxy for broadly available pretraining data, while synthetic nonexistent entities test behavior on information expected to lie outside it. The datasets cover distinct response formats: PreciseWikiQA evaluates short answers, LongWiki evaluates long-form generation, and NonExistentRefusal evaluates whether the model abstains when queried about nonexistent entities. We follow the HalluLens procedure [4]. For

²<https://huggingface.co/datasets/wandb/RAGTruth-processed>

Dataset	Train + eval queries	Avg. document length	Avg. target length
PRECISEWIKIQA	1000 + 2000	37.9	5.8
LONGWIKI	1000 + 2000	38.4	235.0
NONEXISTENTREFUSAL	1000 + 2000	17.6	—
RAGTRUTH	1000 + 1960	698.2	194.4*

Table 1: Summary statistics of the benchmark datasets. Token counts with Llama-2’s tokenizer. *For RAGTRUTH, the reported target length corresponds to the average generated response length. NONEXISTENTREFUSAL has no reference target answer because the desired behavior is abstention.

PreciseWikiQA and LongWiki, examples are built from sampled Wikipedia pages: a question and a gold answer are generated from the page. The gold answer is the response to the question, which serves as a reference for evaluation. The evaluated models then produce responses to these questions. Because the datasets do not provide direct human hallucination annotations, the same LLM³ is used as judge: for PreciseWikiQA and LongWiki, it evaluates model responses against the gold answers; for NonExistentRefusal, it evaluates whether the response is a correct abstention. We therefore treat the three datasets as separate evaluation tasks, each with its own response-level target (refer to Bang et al. [4] for more details).

For PreciseWikiQA, $Q(x_j, y_j)$ is binary answer correctness relative to the gold answer. For LongWiki, the judge decomposes both the gold answer and the model response into atomic claims. Precision is the proportion of response claims supported by the gold answer, and recall is the proportion of gold-answer claims recovered by the response; quality is their harmonic mean. Following Bang et al. [4], claim counts are capped at $K = 32$, yielding the $Q(x_j, y_j) = F1@K(x_i, y_j)$ target.

3.2 Choice of models and computation of uncertainty estimators

Models. The model selection is constrained by RAGTruth, where each query is associated with six model-generated responses annotated by humans for hallucination [37]. Among these models, the three open-weight models are retained: Mistral-7B-Instruct-v0.2, Llama-2-7b-chat-hf, and Llama-2-13b-chat-hf. This choice ensures token-logit access and improves reproducibility under a controlled evaluation protocol.

Recovering token logits for annotated RAGTruth responses. Because RAGTruth provides responses generated during dataset construction and later annotated for hallucinations, evaluation must be performed on these annotated responses. Regenerating responses would break the correspondence with the original annotations. However, token logits are not provided and must be recovered for information-based and white-box sample-based estimators, which require token-level probabilities on the provided generation.

In our study, we recover these probabilities by scoring each annotated response under the corresponding causal language model with teacher forcing [53]. Given an input–response pair (x, y) , we run the model on the concatenated sequence $x \oplus y$ and extract logits at the response positions. This yields the autoregressive conditionals $p_\theta(y_t | x, y_{<t})$, i.e., the probability assigned to each observed token given the input and observed prefix. These probabilities are then used by uncertainty estimators requiring token-level logits for the annotated generation.

Dataset splits for training-based uncertainty estimators. Training-based estimators require training data. Following Vashurin et al. [47], RAGTruth is partitioned at the query level into 1000 training and 2000 evaluation queries, ensuring that all responses associated with a query remain in the same split. For HalluLens, 3000 question-answer pairs are constructed following Bang et al. [4] and split using the same 1000/2000 train-evaluation ratio. All uncertainty estimators are evaluated on the same 2000 evaluation instances; the 1000-instance training split is used only by training-based estimators requiring fitting or training-set statistics. Summary statistics are reported in Table 1.

Auxiliary reference-based score: AlignScore. We also report AlignScore [58] as an auxiliary comparison score. It is reference-based because it evaluates a response by comparing it to an external reference text, such as a gold answer, a source Wikipedia page, or an abstention template. AlignScore is not an uncertainty estimator but an encoder-based model fine-tuned to estimate whether

³hugging-quants/Meta-Llama-3.1-70B-Instruct-AWQ-INT4

one text is supported by another. For PreciseWikiQA and LongWiki, $\text{AlignScore}(y, \text{ref})$ measures whether claims in the model response are supported by the gold answer or source Wikipedia page, yielding a precision-like support signal. The reverse score, $\text{AlignScore}(\text{ref}, y)$, measures whether claims in the reference are recovered by the response, yielding a recall-like recovery signal. For NonExistentRefusal, where no gold answer is available, responses are compared against canonical abstention templates (Section C.1). We use AlignScore to assess the strength of the association between these reference-based signals and the response quality measured by Q . This comparison also helps indicate whether a dataset target is closer to a precision-like support measure, a recall-like recovery measure, or a mixture of both, while making explicit that AlignScore uses reference information unavailable to uncertainty estimators.

4 Evaluation methodology

Let $\mathcal{D} = \{(x_j, y_j)\}_{j=1}^n$ denote an evaluation set, where x_j is the input and y_j is the model response. Each dataset defines a response-level quality target $\{Q(x_j, y_j)\}_{i=1}^n$, as described in Section 3.1. Each uncertainty estimator produces a scalar score u_j for the same instance. Our evaluation assesses whether larger uncertainty values are associated with lower hallucination-specific quality. Uncertainty scores are computed with LM-POLYGRAPH [47] and evaluated along four axes: discrimination, selective prediction, rank calibration, and redundancy.

All reported metrics are accompanied by bootstrap estimates of statistical variability. For each dataset-model pair, 1,000 bootstrap replicates are generated from the evaluation set, each metric is computed on every replicate, and the resulting standard deviation is reported.

4.1 Evaluating hallucination detection

Discriminating hallucinated from supported responses. The first evaluation axis tests whether uncertainty separates lower-quality or hallucinated generations from higher-quality ones. For datasets with binary response-level targets (RAGTruth, PreciseWikiQA, NonExistentRefusal), AUROC is reported with lower-quality or hallucinated responses as the positive class. For LongWiki, the claim-based F1@K target is binarized at 0.5 to enable the same AUROC comparison. This binarization is restricted to discrimination analysis; continuous F1@K is retained for selective-prediction analyses.

Selective prediction. A complementary axis evaluates whether uncertainty can support rejection-based quality control. In selective prediction, high-uncertainty instances are removed, and the quality of the retained responses is measured as the rejection rate increases. Let π a permutation of $\{1, \dots, n\}$ denote the ordering induced by the uncertainty scores $\{u_j\}_{i=1}^n$, sorted from lowest to highest uncertainty. For a rejection rate $r \in [0, 1]$, the retained subset is $S(r) = \{\pi_1, \dots, \pi_{\lfloor (1-r)n \rfloor}\}$. The average retained quality is $\mathcal{Q}(r) = \frac{1}{|S(r)|} \sum_{j \in S(r)} Q(x_j, y_j)$, which defines a quality-retention curve $r \mapsto \mathcal{Q}(r)$. The oracle orders instances by decreasing quality, equivalently rejecting the lowest-quality ones first. Following prior works [34, 47], the curve is summarized with the Prediction Rejection Ratio (PRR),

$$\text{PRR} = \frac{\text{AUC}_U - \text{AUC}_{\text{random}}}{\text{AUC}_{\text{oracle}} - \text{AUC}_{\text{random}}},$$

where AUC_U is the area under the quality-retention curve induced by the uncertainty estimator, $\text{AUC}_{\text{random}}$ corresponds to random rejection, and $\text{AUC}_{\text{oracle}}$ corresponds to the optimal ordering induced by the quality target. Higher PRR values indicate that rejecting high-uncertainty instances yields larger quality gains relative to random rejection.

Rank calibration. Discrimination and selective prediction assess separability and rejection utility; rank calibration instead asks whether the uncertainty ordering remains globally aligned with expected quality. Rank-Calibration Error (RCE; 21) is used as a rank-based diagnostic of whether uncertainty scores are monotonically aligned with response quality. RCE allows comparing estimators with different scales, including unbounded, and does not require ad hoc binarization of continuous targets.

4.2 Redundancy and complementarity across estimators

After evaluating individual performance, we assess whether estimators capture distinct or overlapping signals. Redundancy is analyzed at two levels: similarity in instance rankings within each dataset-

model pair (which we refer to as a *panel*), and similarity in performance profiles across datasets and models. Together, these analyses indicate whether uncertainty estimator groups are interchangeable or complementary, and whether heterogeneity is driven primarily by tasks or models.

Estimator-level correlation. For each panel (dataset-model pair), we compute pairwise Spearman correlations between estimator scores to measure how similarly two estimators rank instances. The resulting correlation matrices are aggregated and visualized with hierarchical clustering to identify groups of estimators with similar behavior.

Performance-level correlation. Score-level similarity within a panel does not show whether estimators succeed in alternative evaluation settings. We therefore compare estimator performance profiles across panels. For each estimator, AUROC and PRR values are collected across all dataset-model pairs, and Kendall’s τ is computed between these profiles to assess whether the estimators exhibit similar performance patterns across tasks and models.

5 Results

We report results on four datasets covering different hallucination types, RAGTruth (RT), PreciseWikiQA (PWQA), LongWiki (LW), and NonExistentRefusal (NR), and three models: Mistral-7B-Instruct-v0.2 (M7B), Llama-2-7B-chat (L7B), and Llama-2-13B-chat (L13B). This yields 12 dataset-model pairs, referred to as *panels*. Unless stated otherwise, estimator rankings exclude AlignScore variants, which are used as reference-based comparators. AUROC, prediction-rejection ratio (PRR), and rank-calibration error (RCE) induce similar estimator rankings across panels (median Spearman correlation +0.97 for AUROC-PRR; −0.96 for AUROC-RCE, where the negative sign reflects that lower RCE is better; Table 8). We therefore use AUROC as the primary discrimination metric, and refer to PRR and RCE only when they add distinct information. Our empirical results, detailed below, provide insights on the relevance of uncertainty estimators for LLM hallucination by assessing their discriminative power, the heterogeneity of results across hallucination types and whether the best estimators provide complementary or redundant signal.

Average performance and consistency across panels. Figure 1 compares each estimator’s mean AUROC across panels with the variability of its within-panel rank. Estimators in the upper part of the plot discriminate hallucinations more strongly on average; estimators on the left rank more consistently across panels. CCP (logit/information-based) and CocoaMSP (white-box sample-based) provide the best compromise between the two criteria. MSP (logit/information-based) and AttentionScore (internal-state) achieve comparable or higher mean AUROC but exhibit greater rank variability, indicating that their averages are driven by panels where they peak rather than by uniform performance across panels. Training-based density estimators (Mahalanobis Distance, MD; Relative MD; Robust Density Estimation, RDE) and the information-based pointwise mutual information variants (PMI, CPMI) recur near the bottom of the plot. The pooled view, therefore, gives the main warning: uncertainty is informative, but no estimator is uniformly reliable. In the following, we show that the variability in Figure 1 is driven primarily by the dataset and secondarily by the model.

The dataset is the primary source of heterogeneity in estimator rankings. Estimator rankings vary more with dataset changes than with model changes. This pattern is reasonable in hindsight, and thus reassuring, but it was not trivial a priori. The Kendall’s τ heatmaps in Figures 5 and 6 (see appendix) quantify this pattern: when the model is fixed and the dataset changes, mean estimator-ranking agreement is low ($\tau = 0.10$ to 0.15); when the dataset is fixed and the model changes, agreement is higher but uneven, with stronger transfer on LW ($\tau = 0.71$) and PWQA ($\tau = 0.62$) than on RT ($\tau = 0.35$) and NR ($\tau = 0.30$).

We investigate this heterogeneity further by comparing the behavior of the estimators across hallucination types. Figure 2 reports family-level predictive performance of uncertainty estimators for two contrasting hallucination settings: PWQA, where most estimators exhibit moderate to strong discriminative power, and RT, where it is sparsest (see Section 2.2 for the family definitions). On PWQA, all six uncertainty estimator families lie above the 0.5 baseline of an uninformative score, with the information-based and white-box sample-based families reaching the highest family-aggregate AUROCs. Hence, our results indicate that uncertainty estimators provide useful discriminative power for hallucination on PWQA. In contrast, family-level ROC curves on RT sit closer to the 0.5 baseline, suggesting that uncertainty estimators carry limited information for discriminating hallucinations in this setting. The two remaining datasets, LW and NR, are reported in Figure 7 (see appendix).

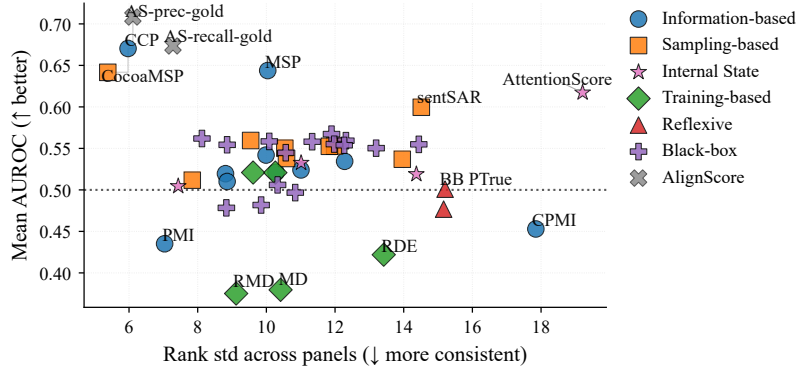


Figure 1: Mean AUROC against rank variability across the 12 panels. Each point represents one uncertainty estimator or AlignScore variant. Rank variability is the standard deviation of an estimator’s within-panel rank across panels; lower values indicate a more stable cross-panel ranking. AlignScore variants are reference-based comparators.

Model differences still affect which estimators perform best. Although task differences dominate overall, the model still changes which uncertainty signal is most informative on a given dataset. On RT with M7B, the leading estimators include the information-based MSP and CCP, as well as the white-box, sample-based CocoaMSP (see Table 4). This pattern does not persist on the Llama models. On RT with L13B, by contrast, reflexive estimators, in which the language model is prompted to assess its own uncertainty (e.g., PTrue), perform best, whereas they are uninformative on the other models. Thus, even within a fixed task setting, estimator choice remains model-dependent.

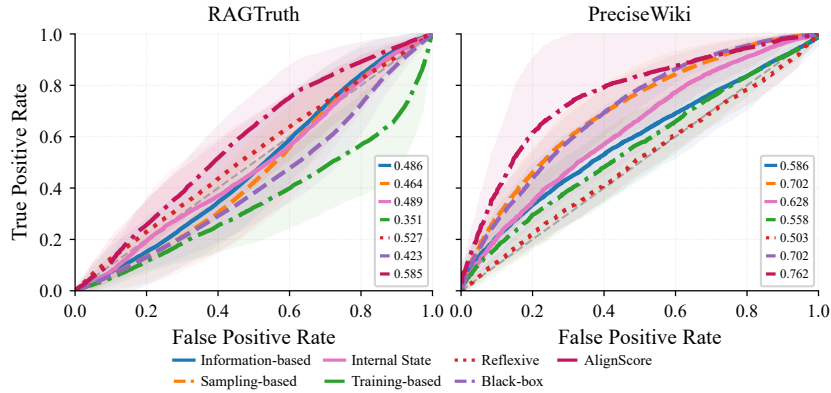


Figure 2: Family-level ROC aggregates per task, averaged across the three generators. For each task–model panel, estimator ROC curves are first averaged within each family by taking the mean true positive rate at each false positive rate. For each task, these model-level family means are then averaged across models. Shaded bands show one standard deviation of these model-level family-mean true positive rates at each false positive rate; they reflect cross-model variability.

Some families consistently perform poorly. The bottom of the panel ranking in Figure 1 is more stable than the top across datasets and models (see Figure 4 in the appendix). The training-based density estimators (MD, Relative MD, RDE) are consistently among the bottom five in nearly every panel, with the information-based PMI and CPMI variants following closely. A plausible interpretation is that density-based estimators measure how typical a generated response is under a fitted reference representation distribution. Since hallucinated responses are still sampled from the generator, they need not be atypical in representation space. Performance may also depend on modeling choices such as the training data, background corpus, representation layer and dimensionality reduction. These results suggest that training-based density estimators and PMI/CPMI should not be used as default uncertainty signals without dataset-specific validation.

The recurrent top estimators form three correlated ranking clusters. Figure 3 reports Spearman’s ρ correlations between the 7 estimators with the highest mean AUROC after averaging across panels. This figure shows that three correlated ranking clusters emerge. The first cluster is logit- and sample-based, and relies on logits: SAR, Maximum Sequence Probability (MSP), Claim-Conditioned Probability (CCP), and CocoaMSP. The strongest correlations occur between MSP, CCP, and CocoaMSP ($\rho \geq 0.78$), while SAR is more weakly attached to the same cluster. This correlation is informative because these estimators differ in construction: MSP is a direct sequence-likelihood score; CCP filters local token alternatives through a natural-language inference model; CocoaMSP modulates an MSP-derived score using inter-sample semantic dissimilarity; and SAR combines token-level and sentence-level relevance signals across sampled responses. The second cluster contains the two Eccentricity variants, based respectively on contradiction (Ecc-c) and entailment (Ecc-e) relations. Both are black-box graph-based estimators: they build a semantic-relation graph over sampled responses and quantify dispersion from the resulting graph spectral representation. Their moderate correlation ($\rho = 0.36$) suggests that entailment- and contradiction-based graph constructions are related but not interchangeable. The third cluster is a singleton, AttentionScore, which uses internal-state representations rather than logits or generated samples. This likely explains why its correlations with the other estimators remain low.

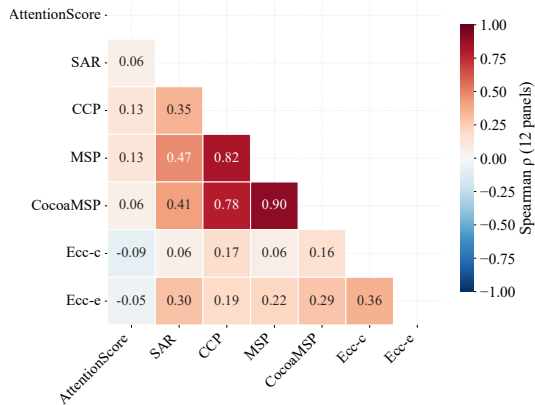


Figure 3: Spearman correlation between the top uncertainty estimators across panels.

Access constraints should shape estimator selection. The ranking clusters differ not only in performance, but also in input requirements. The first cluster mostly contains white-box estimators that require token probabilities. CocoaMSP and SAR further require sampled generations and a sentence-similarity model, while AttentionScore requires access to attention weights. In contrast, the NLI-graph estimators are black-box with respect to the generator: they use sampled text responses and an auxiliary NLI model, but do not require generator logits, hidden states, or attention weights. When white-box access is available, CocoaMSP and CCP perform better across datasets, while MSP remains an attractive, computationally efficient baseline.

6 Conclusion

Our study clarifies when existing uncertainty estimation methods align with hallucination-related output quality in large language models. The main finding is that uncertainty is informative, but only conditionally. Task-specific evaluation targets are the main source of heterogeneity, which argues against treating any uncertainty estimator as a universal hallucination detector. Instead, uncertainty estimators should be selected for practical use based on the target task, the level of access to the generator, the computational budget, and validation performance on the target setting. The benchmark also suggests a practical structure for estimator selection: (i) black-box Eccentricity variants are strong candidates when logits are unavailable, (ii) logit-based estimators such as MSP, CCP, and related methods provide a strong default family when white-box access is available, and (iii) AttentionScore appears most useful in context-faithfulness and abstention tasks.

Limitations and future work

There are also a few limitations to our study. There are also a few limitations to our study. Each dataset determines whether a response is hallucinated according to its own annotation or reference protocol. In particular, LongWiki uses an F1 target that is a long-form quality signal rather than a pure hallucination label. Class imbalance, that is, variation across models in the proportion of responses labeled as hallucinated, may also affect panel-level comparisons, although bootstrap intervals partly quantify sampling variability. Finally, the evaluated model set is controlled and reproducible but does

not cover larger frontier systems or the full range of post-training regimes. The influence of other factors, such as the different phases of the training paradigm, could also be worth investigating.

Future work could extend this evaluation in three directions. First, hallucination targets should be made more fine-grained and stratified. Second, localization-aware analyses should use span-level annotations, such as those available in RAGTruth, to test whether uncertainty scores indicate where hallucinations occur in the answer, rather than only whether a response is unreliable.

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A Related works

A.1 Evaluating uncertainty estimation methods

Work on evaluating uncertainty estimation (UE) methods for LLMs has addressed three concerns: building infrastructure that places heterogeneous methods on a common framework; assessing whether comparisons drawn under controlled conditions hold up in deployment; and questioning whether the evaluation protocols themselves are sound. We add a fourth: whether the evaluation *target* – correctness as commonly operationalized – adequately reflects the phenomenon (hallucination) that uncertainty is supposed to detect.

Method-coverage frameworks. The most direct response to method heterogeneity has been to consolidate estimators behind shared interfaces and shared intermediate quantities. LM-POLYGRAPH [47] implements a comprehensive estimator suite behind a single pipeline that computes greedy responses, token probabilities, sampled responses, and semantic relation matrices once and reuses them across estimators, controlling for implementation differences. Adjacent toolkits – UQLM [6], UNCERTAINTYZOO [54], and UBENCH [51] – provide complementary coverage or standardize estimator responses to a common confidence range. Across these frameworks, the contribution is comparability rather than validity: each evaluation is only as meaningful as its target for correctness.

Deployment- and protocol-level critiques. A second strand asks whether rankings produced under controlled benchmarks transfer to deployment, and whether the protocols themselves can be trusted. Bakman et al. [3] evaluate nineteen UE methods under decision-threshold drift, prompt perturbation, long-form adaptation, and ensembling, and find that most methods are sensitive to threshold choice and vulnerable to adversarial prompts – patterns invisible under standard evaluation conditions. Santilli et al. [42] show that the same nuisance factor can bias both UE scores and correctness functions – most strikingly response length – so that a trivial length baseline can rival principled methods under length-biased correctness metrics. Ielanskyi et al. [22] reinforce this concern by showing that the choice of approximate correctness function (ROUGE, BLEURT, LLM-as-judge variants) substantially reorders method rankings, and propose marginalising over multiple judges as a more robust alternative. Complementary critiques target the limited ecological validity of QA-style evaluations [12], the inability of realisable benchmarks to test calibration over inherently variable outcomes [10, 45], and the incomparable scales on which estimator responses live [21]. The shared message is that “method *A* outperforms method *B* on benchmark *X*” is routinely confounded by length, choice of correctness function, threshold calibration, or estimator response scale – and that fixing the benchmark itself does not fix the comparison.

The evaluation gap we address. Across these strands, a common limitation is that the evaluation *target* is rarely a hallucination-specific signal: correctness is operationalized through lexical or semantic overlap with a reference answer, or through a multiple-choice answer key, rather than through an explicit characterization of when an response is unsupported by the evidence the model was supposed to use. Our contribution sits in this space: we retain the LM-POLYGRAPH estimator catalog and the methodological caution of Santilli et al. [42], Bakman et al. [3], and Ielanskyi et al. [22], but replace generic correctness targets with hallucination targets defined through operational distinctions established in the hallucination literature, which we develop next.

A.2 Hallucination: definitions and benchmarks

The definition of hallucination has shifted as LLMs have moved from text-conditioned to largely open-ended generation, and the choice of definition directly shapes what a benchmark measures.

From NLG taxonomy to LLM-specific definitions. The intrinsic/extrinsic distinction was originally formulated for task-conditioned NLG by Ji et al. [23]: an response is *intrinsically* hallucinated if it contradicts an input source, and *extrinsically* hallucinated if it cannot be verified from that source – a notion that presupposes a single explicit input. As LLMs increasingly generate free-form text without such a source, two reframings followed. Huang et al. [20] proposed *factuality* versus *faithfulness* as the LLM-era replacement axis (correctness against the world vs. adherence to context), while Augenstein et al. [2] argued that *factuality* and *hallucination* should be treated as distinct concepts entirely: factuality requires an external oracle, whereas hallucination admits an internal definition – whether the response matches the evidence the model was supposed to use. The separation is consequential because the two concepts overlap but are not identical: a model can be factually correct while unfaithful to its evidence, or factually wrong while faithful to a misleading source.

HalluLens and the operational extrinsic/intrinsic split. Bang et al. [4] consolidate this picture and introduce the HALLULENS benchmark. They redefine *intrinsic* hallucinations as responses unsupported by information *explicitly provided at inference time*, and *extrinsic* hallucinations as responses unsupported by the *pretraining knowledge the model is expected to have encountered*. The reformulation is operational: each regime corresponds to a concrete source of evidence against which an response can be adjudicated, with ground-truth labels defined without implicit appeal to world truth. HALLULENS instantiates this with three extrinsic tasks anchored to Wikipedia, along with items generated on-the-fly to mitigate test-set leakage. The Wikipedia anchor rests on a documented approximation – recent pretraining mixtures are assumed to contain Wikipedia, so inconsistency with a Wikipedia article is treated as a proxy for inconsistency with pretraining knowledge – imperfect for long-form generation [16] but the most operationalized proxy currently in use.

Benchmark heterogeneity and our positioning. With the operational split in hand, the existing landscape can be repartitioned by the definition each benchmark encodes. *Explicitly intrinsic* benchmarks include RAGTRUTH [37], which provides word-level hallucination annotations for retrieval-augmented generation. Closed-book QA benchmarks such as TRIVIAQA [24], NATURAL QUESTIONS [28], and SIMPLEQA [52] target some dimension of factuality but sit ambiguously with respect to the operational split: classifying them as extrinsic-regime benchmarks would require evidence that all their questions are answerable from the model’s pretraining data, which is generally not established. HALLULENS fills this gap by restricting questions to those generated from Wikipedia, making the pretraining-coverage assumption a documented approximation rather than an implicit one; *explicitly extrinsic* benchmarks otherwise remain comparatively rare. As a result, comparing performance across these benchmarks partly compares definitions rather than methods. We therefore pair RAGTRUTH (intrinsic regime; word-level labels produced before the operational split was formalized, hence not biased toward any one definitional choice) with the three HALLULENS tasks – PRECISEWIKIQA, LONGWIKI, and NONEXISTENTREFUSAL – as the most explicit operational extrinsic benchmarks currently available with controlled test-set generation. Together, these commit us to the hallucination/factuality distinction articulated by Augenstein et al. [2] and operationalized by Bang et al. [4], and yield measurements that remain comparable across regimes without conflating them.

B Description and implementation details on uncertainty estimators

This appendix specifies the uncertainty estimators evaluated in our benchmark, briefly described in Section 2. We pursue two goals jointly: a conceptual map organized by the signal each method exploits, and an implementation-faithful description of the exact variants executed in our pipeline. Every formula below was derived by auditing the corresponding implementation in our LM-POLYGRAPH-based pipeline [15, 47] rather than from the high-level descriptions in the original papers.

Pipeline. We use LM-POLYGRAPH as the uncertainty-score computation engine. The framework separates two stages: shared intermediate quantities (greedy responses, token log-probabilities, stochastic samples, semantic relation matrices, internal representations) are computed *once* per input, and the estimators consume these quantities to produce the final scores. We do not modify any estimator implementation. The pinned version of LM-POLYGRAPH we use is [main/0e5bccdd35efe7de4e3ae8124427aa12575a1bd2c](https://github.com/LLM-POLYGRAPH/LLM-POLYGRAPH/tree/main/0e5bccdd35efe7de4e3ae8124427aa12575a1bd2c). Downstream evaluation is performed in our analysis pipeline, externally to LM-POLYGRAPH.

B.1 Implementation details

Table 2 consolidates the numerical hyperparameters and external models used in the executed pipeline. Default LM-POLYGRAPH settings are used for any value not listed.

B.2 Code availability

The code is available here: <https://anonymous.4open.science/r/uncertainty-benchmark-867C>.

B.3 Compute resources

All GPU-demanding computation was performed on a university HPC cluster using usage-constrained GPU allocations. The primary hardware was an NVIDIA RTX PRO 6000 Blackwell (96 GB GDDR7) for large-model runs, with NVIDIA RTX 6000 Ada/RTX A6000 (48 GB GDDR6) nodes used for smaller models. Nodes were equipped with 24–32 CPU cores and 192–512 GB RAM. The two main GPU-demanding phases totaled approximately 700 productive GPU-hours:

- Benchmark inference (3 tasks $\times$ 3 models, up to 2,000 samples each, 50 uncertainty estimators): roughly 415 GPU-hours for HalluLens tasks and 255 GPU-hours for RAGTruth, ranging from 5 GPU-hours per run (small datasets, 7B models) to 87 GPU-hours (large datasets, 13B models). 48 GB of VRAM suffices for 7B models at a batch size of 1; 13B models require 96 GB.
- LongWiki evaluation (Llama-3.1-70B-AWQ judge via vLLM): 31 productive GPU-hours across 2 model runs, using 2 GPUs simultaneously (judge server + evaluation client).

B.4 Notation and conventions

Let x denote the input query and $\hat{y} = (\hat{y}_1, \dots, \hat{y}_T)$ the *response to be scored* of length T . In the standard free-generation setting, $\hat{y}$ is the greedy response of the model; in the RAGTruth dataset, where responses have already been generated and annotated, $\hat{y}$ is the observed response supplied by the dataset. The autoregressive conditional probability of the token $\hat{y}_t$ is $p_\theta(\hat{y}_t \mid x, \hat{y}_{<t})$, and the unconditional language-model probability (input x removed) is $p_\theta(\hat{y}_t \mid \hat{y}_{<t})$. The full predictive distribution at position t lives on the simplex $\Delta(\mathcal{V})$ over the vocabulary of size $V = |\mathcal{V}|$, i.e. $p_\theta(\cdot \mid x, \hat{y}_{<t}) \in \Delta(\mathcal{V})$. The full sequence probability is $P_\theta(\hat{y} \mid x) = \prod_{t=1}^T p_\theta(\hat{y}_t \mid x, \hat{y}_{<t})$. Shannon entropy of any $q \in \Delta(\mathcal{V})$ is $\mathcal{H}(q) = -\sum_{v \in \mathcal{V}} q(v) \log q(v)$, and $\mathcal{U}(\mathcal{V})$ denotes the uniform distribution on $\mathcal{V}$.

When stochastic sampling is required, we draw $S = 10$ responses $y^{(1)}, \dots, y^{(S)} \sim p_\theta(\cdot \mid x)$ at sampling temperature 1 with random seed 42. We write $|y^{(s)}|$ for the length of sample s and $P_\theta(y^{(s)} \mid x) = \prod_{t=1}^{|y^{(s)}|} p_\theta(y_t^{(s)} \mid x, y_{<t}^{(s)})$ for its sequence probability. The same sample pool is shared across all sample-based estimators for a given input, so any two such estimators are compared on identical samples. For semantic-consistency methods, NLI-based pairwise scores are

$$E_{ij} = p_{\text{NLI}}(\text{entailment} \mid y^{(i)}, y^{(j)}), \quad C_{ij} = p_{\text{NLI}}(\text{contradiction} \mid y^{(i)}, y^{(j)}),$$

produced by `microsoft/deberta-large-mnli` [19].⁴ Cross-encoder semantic similarity, when used, is computed with `cross-encoder/stsb-roberta-large`.⁵

Score orientation. All estimators are reported as *uncertainty* scores: larger values indicate greater uncertainty. Two estimators in LM-POLYGRAPH are confidence-oriented (Rényi Divergence and Fisher–Rao Distance) and are negated before analysis. The formulas below are uncertainty-oriented.

Name versus executed formula. Two estimator names refer to a probability, whereas the executed scores are log-transformations of that probability. We make this explicit to avoid confusion. Maximum Sequence Probability refers to the quantity $P_\theta(\hat{y} \mid x)$, but the implemented score in LM-POLYGRAPH is its negative log – the sequence negative log-likelihood (NLL). Similarly, Perplexity refers to $\exp(-\frac{1}{T} \sum_t \log p_\theta(\hat{y}_t \mid x, \hat{y}_{<t}))$, whereas the executed score is the *logarithm* of this perplexity, i.e. the length-normalized sequence negative log-likelihood (LN-NLL).

⁴<https://huggingface.co/microsoft/deberta-large-mnli>

⁵<https://huggingface.co/cross-encoder/stsb-roberta-large>

Definitional convention. We use $:=$ for the exact quantity returned by the implementation and $=$ for subsequent algebraic rewrites.

Configuration convention. Unless stated otherwise, we use the default LM-POLYGRAPH configuration. When a hyperparameter materially affects the score, we report its value, see Table 2.

B.5 White-box estimators

White-box estimators require access to information beyond the generated text of the response: the model’s logits, attention maps, hidden states, or training data. We organize them into five families, ordered from the most direct readouts of the predictive distribution to the most indirect: (i) information-based estimators, which aggregate token-level distributional quantities computed in a single forward pass; (ii) sample-dispersion estimators, which lift these quantities to multiple stochastic samples and combine them with measures of inter-sample dispersion; (iii) internal-state estimators, which use attention or hidden-state geometry; (iv) training-based estimators, which score representation-space distance from a fitted reference distribution; and (v) reflexive estimators, which prompt the model to judge its own answer.

B.5.1 Information/Logit-based estimators

These estimators consume only the token-level predictive distributions encountered along a single response $\hat{y}$ and aggregate a position-wise quantity across the sequence. Within this family we distinguish methods that look only at the probability of the realized token (Maximum Sequence Probability, Perplexity), methods that examine the full predictive distribution at each position (Mean Token Entropy, SelfCertainty, Rényi Divergence, Fisher–Rao Distance), methods that contrast the conditional predictive distribution with its unconditional counterpart (Pointwise Mutual Information and its conditional variant), and two methods that introduce auxiliary signals at the token level (TokenSAR, Claim-Conditioned Probability).

Maximum Sequence Probability. Maximum Sequence Probability is the most basic logit-based estimator: it considers only the probability assigned to the realized response and ignores the rest of the predictive distribution. The executed score is the sequence negative log-likelihood,

$$U_{\text{MSP}}(x; \hat{y}) := - \sum_{t=1}^T \log p_{\theta}(\hat{y}_t \mid x, \hat{y}_{<t}) = - \log P_{\theta}(\hat{y} \mid x).$$

The score is zero when the model assigns probability 1 to every realized token and grows without bound as confidence in the realized response drops. It is length-sensitive: longer responses accumulate more uncertainty at constant per-token confidence.

Perplexity. Perplexity removes the length sensitivity of Maximum Sequence Probability by averaging the per-token negative log-likelihood,

$$U_{\text{PPL}}(x; \hat{y}) := - \frac{1}{T} \sum_{t=1}^T \log p_{\theta}(\hat{y}_t \mid x, \hat{y}_{<t}),$$

which equals the logarithm of the standard perplexity of $\hat{y}$ under the model. Like Maximum Sequence Probability, this score uses only the probability of the realized token, but it is comparable across responses of different lengths [18].

Mean Token Entropy. Mean Token Entropy is the simplest logit-based estimator that uses the *full* predictive distribution at each position rather than only the probability of the realized token. The score averages the Shannon entropy of the predictive distribution across positions, which equals the chain-rule expansion of the sequence entropy [9, 34]:

$$U_{\text{MTE}}(x) := \frac{1}{T} \sum_{t=1}^T \mathcal{H}(p_{\theta}(\cdot \mid x, \hat{y}_{<t})).$$

Unlike MSP and PPL, MTE is not determined by the realized token alone: even when $\hat{y}_t$ is itself among the most probable tokens, MTE can still be large if probability mass is spread broadly across

alternatives; conversely, when $\hat{y}_t$ has only moderate probability but the distribution is peaked on a few competing tokens, MTE remains small because the distribution is concentrated.

SelfCertainty. SelfCertainty compares a uniform reference distribution to the model’s predictive distribution, using the *reverse* KL direction relative to the entropy-like divergence-from-uniform scores used by Rényi Divergence and Fisher–Rao Distance below. Let $\tilde{\mathcal{V}}_t \subseteq \mathcal{V}$ be the set of vocabulary entries with finite log-probability at position t . The implemented token-level quantity is

$$D_{\text{KL}}(\mathcal{U}(\tilde{\mathcal{V}}_t) \parallel p_\theta(\cdot \mid x, \hat{y}_{<t})) = -\log |\tilde{\mathcal{V}}_t| - \frac{1}{|\tilde{\mathcal{V}}_t|} \sum_{v \in \tilde{\mathcal{V}}_t} \log p_\theta(v \mid x, \hat{y}_{<t}),$$

and the sequence-level uncertainty score reverses the polarity by taking the negative mean:

$$U_{\text{SC}}(x) := -\frac{1}{T} \sum_{t=1}^T D_{\text{KL}}(\mathcal{U}(\tilde{\mathcal{V}}_t) \parallel p_\theta(\cdot \mid x, \hat{y}_{<t})).$$

A flat predictive distribution yields a small divergence and $U_{\text{SC}} \approx 0$ (high uncertainty); a sharply peaked distribution yields a large divergence and $U_{\text{SC}} \ll 0$ (low uncertainty). Unlike entropy or Rényi-style scores, the reverse-KL direction is especially sensitive to tokens assigned very small probability: a single very small $p_\theta(v \mid \cdot)$ blows up one term of the sum.

Rényi Divergence. Rényi Divergence also compares the predictive distribution to a uniform reference, but in the *opposite direction* from SelfCertainty – and on a temperature-scaled version of the distribution. Let

$$q_t(v \mid x, \hat{y}_{<t}; \tau) = \text{softmax}(\log p_\theta(\cdot \mid x, \hat{y}_{<t}) / \tau)(v) \in \Delta(\mathcal{V}).$$

The token-level Rényi divergence of order α from q_t to the uniform distribution is

$$D_\alpha(q_t(\cdot \mid x, \hat{y}_{<t}; \tau) \parallel \mathcal{U}(\mathcal{V})) = \frac{1}{\alpha - 1} \log \sum_{v \in \mathcal{V}} q_t(v \mid x, \hat{y}_{<t}; \tau)^\alpha - \log |\mathcal{V}|,$$

and the sequence-level uncertainty score, after polarity reversal, is

$$U_{\text{Rényi}}(x; \tau) := -\frac{1}{T} \sum_{t=1}^T D_\alpha(q_t(\cdot \mid x, \hat{y}_{<t}; \tau) \parallel \mathcal{U}(\mathcal{V})).$$

Larger divergence corresponds to a sharper, more concentrated distribution, hence the negation. The parameter α controls sensitivity to the distribution: $\alpha < 1$ gives more weight to low-probability tokens, $\alpha > 1$ emphasizes the peak; the temperature τ smooths the distribution before the divergence is computed. SelfCertainty should *not* be read as the $\alpha = 1$ case of this estimator: it uses the KL direction $\mathcal{U} \parallel p$, whereas this Rényi score uses $q \parallel \mathcal{U}$, which measures a different geometric quantity (peak concentration rather than coverage). We use $\alpha = 0.5$ and $\tau = 2$ [11].

Fisher–Rao Distance. Fisher–Rao Distance also measures how far the temperature-scaled predictive distribution $q_t(\cdot \mid x, \hat{y}_{<t}; \tau)$ lies from the uniform distribution on $\Delta(\mathcal{V})$, but uses Fisher–Rao geometry rather than a divergence. The token-level geodesic distance is

$$d_{\text{FR}}(q_t(\cdot \mid x, \hat{y}_{<t}; \tau), \mathcal{U}(\mathcal{V})) = \frac{2}{\pi} \arccos \left(\sum_{v \in \mathcal{V}} \sqrt{q_t(v \mid x, \hat{y}_{<t}; \tau) \cdot \frac{1}{|\mathcal{V}|}} \right),$$

and the sequence-level uncertainty score, after polarity reversal, is

$$U_{\text{FR}}(x; \tau) := -\frac{1}{T} \sum_{t=1}^T d_{\text{FR}}(q_t(\cdot \mid x, \hat{y}_{<t}; \tau), \mathcal{U}(\mathcal{V})).$$

The geodesic distance is zero when q_t is uniform and approaches 1 as q_t becomes more peaked; the negation orients the score so that a flatter (more uncertain) distribution receives a larger value. Unlike any divergence, Fisher–Rao is a true metric on the simplex (symmetric, satisfies the triangle inequality). We use $\tau = 2$ [11].

Mean Pointwise Mutual Information. The estimators above measure either how confident the model is in the realized response (MSP, PPL) or how broadly its predictive distribution spreads (MTE, SelfCertainty, Rényi, Fisher–Rao). They cannot detect responses that are likely under the model regardless of the input, that is, responses whose probability does not benefit from prompt grounding. Mean Pointwise Mutual Information addresses this by contrasting, for each realized token, the conditional log-probability $\log p_\theta(\hat{y}_t \mid x, \hat{y}_{<t})$ with the unconditional log-probability $\log p_\theta(\hat{y}_t \mid \hat{y}_{<t})$. The latter is obtained by an additional forward pass on the response alone, with the prompt x removed. The executed score is

$$U_{\text{PMI}}(x; \hat{y}) := -\frac{1}{T} \sum_{t=1}^T [\log p_\theta(\hat{y}_t \mid x, \hat{y}_{<t}) - \log p_\theta(\hat{y}_t \mid \hat{y}_{<t})].$$

When the conditional probability exceeds the unconditional, the prompt is informative and U_{PMI} is negative (low uncertainty); when the two are comparable, the response could have been generated without the prompt and U_{PMI} approaches zero (high uncertainty), which is interpreted as a signal of weak grounding [46].

Mean Conditional Pointwise Mutual Information. Mean Conditional Pointwise Mutual Information refines PMI by gating the unconditional correction: the correction is applied only at positions where the predictive distribution is locally uncertain, and is ignored at sharp positions, where the unconditional log-probability would add noise rather than signal. “Locally uncertain” is operationalized through the same per-position entropy used by MTE: the correction is applied only when $\mathcal{H}(p_\theta(\cdot \mid x, \hat{y}_{<t})) \geq \tau$, with weight λ . The executed score is

$$U_{\text{CPMI}}(x; \hat{y}) := -\frac{1}{T} \sum_{t=1}^T [\log p_\theta(\hat{y}_t \mid x, \hat{y}_{<t}) - \lambda \cdot \mathbf{1}[\mathcal{H}(p_\theta(\cdot \mid x, \hat{y}_{<t})) \geq \tau] \cdot \log p_\theta(\hat{y}_t \mid \hat{y}_{<t})].$$

At low-entropy positions, the score reduces to the realized-token negative log-likelihood (Perplexity-like behavior); at high-entropy positions, it behaves like PMI [46]. We use the LM-POLYGRAPH defaults $\tau = 0.0656$ and $\lambda = 3.599$.

TokenSAR. TokenSAR re-weights the per-token negative log-likelihoods by an external measure of token relevance, so that tokens whose removal changes the meaning of the response contribute more to the score than function words or filler. Token relevance is computed by leave-one-out cross-encoder similarity: for each position t , let $\text{sim}_t \in [0, 1]$ be the semantic similarity between $\hat{y}$ and the response with token $\hat{y}_t$ removed. The unnormalized relevance is $1 - \text{sim}_t$, and the normalized relevance is $R_t = (1 - \text{sim}_t) / \sum_{t'=1}^T (1 - \text{sim}_{t'})$. The executed score is

$$U_{\text{TSAR}}(x; \hat{y}) := \sum_{t=1}^T R_t \cdot [-\log p_\theta(\hat{y}_t \mid x, \hat{y}_{<t})].$$

Only high-relevance improbable tokens drive uncertainty; an improbable but semantically dispensable token has little effect [13].

Claim-Conditioned Probability. Claim-Conditioned Probability targets a different failure mode of standard token-probability estimators: at each position, several alternative tokens may be plausible substitutes, and only some of them would change the meaning of the response. The executed pipeline proceeds in three steps. At each position t , the top-10 token alternatives $\mathcal{A}_t = \{a_{t,0}, a_{t,1}, \dots, a_{t,9}\}$ are identified, where $a_{t,0} = \hat{y}_t$ is the greedy token itself. Each alternative $a_{t,i}$ is then mapped to a candidate response by substituting it at position t in $\hat{y}$, and bidirectional NLI is evaluated between the greedy-substituted response (i.e., $\hat{y}$ itself) and the $a_{t,i}$ -substituted response. The two directional labels are combined into a single label $c_{t,i} \in \{\text{entail}, \text{contra}, \text{neutral}\}$: if both directions agree, that label is used; if one direction is entail and the other contra, the label is neutral; otherwise the single non-neutral label is used. Alternatives with $c_{t,i} = \text{entail}$ and the greedy alternative $a_{t,0}$ (which is trivially entailing) form the entailment set $\mathcal{E}_t$, alternatives with $c_{t,i} = \text{contra}$ form the contradiction set $\mathcal{C}_t$, and neutral alternatives are discarded. At position t , the claim-conditioned probability is the share of probability mass on entailing alternatives within the entailing-or-contradicting subset,

$$\text{CCP}_t = \frac{\sum_{a \in \mathcal{E}_t} p_\theta(a \mid x, \hat{y}_{<t})}{\sum_{a \in \mathcal{E}_t \cup \mathcal{C}_t} p_\theta(a \mid x, \hat{y}_{<t})},$$

and the sequence-level uncertainty score is the negated product across positions,

$$U_{\text{CCP}}(x) := - \prod_{t=1}^T \text{CCP}_t.$$

Since neutral alternatives are excluded from the denominator, the ratio measures semantic support *conditional on the alternative being either entailing or contradicting*, not support against the full predictive distribution. A single position at which probability leaks toward contradicting alternatives is enough to pull the product toward zero and the score toward zero in absolute value (high uncertainty); concentration of mass on entailing alternatives produces a near-1 product and a score near -1 (low uncertainty) [14].

B.5.2 Sample semantic dispersion estimators

Sample-dispersion estimators draw S stochastic samples $y^{(1)}, \dots, y^{(S)}$ from the predictive distribution and combine sample-level probabilities with a measure of dispersion across samples. Beyond the probabilities themselves, several estimators in this family use auxiliary signals (semantic relations from an NLI model, semantic similarity from a cross-encoder, or internal embeddings); we characterize them as probability-aware sample-based estimators rather than as pure consistency estimators. They mirror the structure of the logit-based family: probability-only methods (Monte Carlo Sequence Entropy and its length-normalized variant), methods that summarize agreement under a notion of meaning (Semantic Entropy, Semantic Density), and hybrid methods that combine probability with an external relevance or consistency signal (SentenceSAR, SAR, the Cocoa family).

Monte Carlo Sequence Entropy. Monte Carlo Sequence Entropy approximates the model’s expected sequence negative log-likelihood under its own sampling distribution:

$$U_{\text{MCSE}}(x) := -\frac{1}{S} \sum_{s=1}^S \log P_{\theta}(y^{(s)} | x) = -\frac{1}{S} \sum_{s=1}^S \sum_{t=1}^{|y^{(s)}|} \log p_{\theta}(y_t^{(s)} | x, y_{<t}^{(s)}).$$

A model that consistently produces high-probability samples receives a low score; a model that scatters probability mass across many possible responses receives a high score [34].

Monte Carlo Normalized Sequence Entropy. Monte Carlo Normalized Sequence Entropy is the length-normalized version of the previous estimator: each sample’s negative log-likelihood is divided by its own length before averaging, so that the score is comparable across prompts that elicit responses of different lengths:

$$U_{\text{MCNSE}}(x) := -\frac{1}{S} \sum_{s=1}^S \frac{1}{|y^{(s)}|} \log P_{\theta}(y^{(s)} | x).$$

Semantic Entropy. Semantic Entropy uses the same sample-based negative log-likelihoods, but groups them by meaning before computing entropy, so that paraphrases of the same answer do not inflate the score. The pipeline first partitions the samples into semantic equivalence classes $C_1, \dots, C_M$ using bidirectional NLI entailment: $y^{(i)}$ and $y^{(j)}$ are placed in the same class when they entail one another. The probability of class C_m is then obtained by summing sample probabilities and renormalizing:

$$P(C_m | x) = \frac{\sum_{s: y^{(s)} \in C_m} P_{\theta}(y^{(s)} | x)}{\sum_{s=1}^S P_{\theta}(y^{(s)} | x)}.$$

The uncertainty score is the entropy of this class distribution:

$$U_{\text{SE}}(x) := - \sum_{m=1}^M P(C_m | x) \log P(C_m | x).$$

A model that always produces the same meaning regardless of surface form yields a single class and zero entropy; a model that distributes mass over several meanings yields large entropy [17, 27].

Semantic Density. Semantic Density replaces the hard equivalence classes of Semantic Entropy with a soft NLI-based kernel and a probability-weighted aggregation around the greedy response $\hat{y}$. Let $\tilde{P}_\theta(y^{(s)} | x) = \exp(\frac{1}{|y^{(s)}|} \log P_\theta(y^{(s)} | x))$ denote the length-normalized sample probability, and let K_s denote the NLI-based kernel score that quantifies the soft semantic agreement between sample s and the surrounding samples (including $\hat{y}$). Writing $\tilde{P}_\theta(\hat{y} | x)$ for the length-normalized probability of the greedy response, the executed score is the negated probability-weighted kernel density

$$U_{SD}(x; \hat{y}) := -\frac{\sum_{s=1}^S \tilde{P}_\theta(y^{(s)} | x) K_s + \tilde{P}_\theta(\hat{y} | x)}{\sum_{s=1}^S \tilde{P}_\theta(y^{(s)} | x) + \tilde{P}_\theta(\hat{y} | x)}.$$

The score is close to -1 (low uncertainty) when the probable samples cluster in a dense semantic neighborhood of $\hat{y}$, and close to 0 (high uncertainty) when the kernel scores are uniformly low [39].

SentenceSAR. SentenceSAR extends the relevance-weighting idea of TokenSAR to the multi-sample setting. The contribution of each sample to the score is its own probability augmented by a relevance term that aggregates the probabilities of *other* samples weighted by their semantic similarity to the current one. Writing sim_{sj} for the cross-encoder similarity between $y^{(s)}$ and $y^{(j)}$ and τ for a temperature parameter, the support term is

$$R_S(s; \tau) = \frac{1}{\tau} \sum_{j \neq s} P_\theta(y^{(j)} | x) \text{sim}_{sj},$$

and the executed score is

$$U_{\text{SentenceSAR}}(x; \tau) := \frac{1}{S} \sum_{s=1}^S -\log(P_\theta(y^{(s)} | x) + R_S(s; \tau)).$$

A sample that is both probable and semantically supported by other samples contributes little to the score; a sample that is improbable and semantically isolated contributes a large negative log [13]. We use $\tau = 1$.

SAR. SAR is the full pipeline of Duan et al. [13], combining TokenSAR at the token level and SentenceSAR at the sentence level. The raw sample probability used in SentenceSAR is replaced with the relevance-adjusted probability from TokenSAR. Concretely, with $\tilde{P}_\theta^{\text{tsar}}(y^{(s)}) = \exp(-U_{\text{TSAR}}(x, y^{(s)}))$ the relevance-weighted probability of sample s , the support term becomes $\tilde{R}_S(s; \tau) = (1/\tau) \sum_{j \neq s} \tilde{P}_\theta^{\text{tsar}}(y^{(j)}) \text{sim}_{sj}$, and the executed score is

$$U_{\text{SAR}}(x; \tau) := \frac{1}{S} \sum_{s=1}^S -\log(\tilde{P}_\theta^{\text{tsar}}(y^{(s)}) + \tilde{R}_S(s; \tau)).$$

SAR composes two relevance signals – token-level for each sample and sentence-level across samples – into a single uncertainty score. We use $\tau = 1$.

Cocoa family. The Cocoa family follows a simple template: a base uncertainty score is multiplied by an inter-sample semantic dissimilarity factor, so that the final score is large only when the response is uncertain on its own *and* semantically dissimilar to the alternative samples [48]. Let $u_{\text{base}}(x; \hat{y})$ denote the base uncertainty and let $S^{\text{sent}} \in \mathbb{R}^{K \times K}$ be the cross-encoder sentence-similarity matrix over the K sentence units computed by LM-POLYGRAPH for this input (greedy response and/or samples, depending on the module configuration). The dissimilarity factor is the mean of $1 - S_{ij}^{\text{sent}}$ across the full matrix,

$$D_{\text{sent}}(x) = \frac{1}{K^2} \sum_{i=1}^K \sum_{j=1}^K (1 - S_{ij}^{\text{sent}}),$$

and the executed Cocoa score is the product

$$U_{\text{Cocoa}}(x; \hat{y}) := u_{\text{base}}(x; \hat{y}) \cdot D_{\text{sent}}(x).$$

We instantiate three variants by choosing $u_{\text{base}} \in \{U_{\text{MSP}}, U_{\text{PPL}}, U_{\text{MTE}}\}$, denoted CocoaMSP, CocoaPPL, and CocoaMTE. An improbable response that is semantically consistent with the alternative

samples is treated as a paraphrase and receives a low score; an improbable response that is also semantically isolated receives a high score. *Implementation note:* although the LM-POLYGRAPH source comment states that the diagonal is excluded, the executed code averages over the full $K \times K$ matrix. Since self-similarity is 1, the diagonal entries contribute 0 to $1 - S_{ij}^{\text{sent}}$ and thus dilute D_{sent} by a factor of $(K - 1)/K$.

B.5.3 Internal-state estimators

Internal-state estimators use latent computations of the model – attention weights or hidden representations – rather than only the response distribution. Their common premise is that predictive unreliability may already be visible in the model’s internal processing dynamics before it is fully reflected in the response token distribution.

AttentionScore. AttentionScore measures self-attention concentration along the generated sequence, assuming that tokens with weak self-attention are less stably represented and therefore more error-prone. Writing $\alpha_{\ell,h}(t, t) \in [0, 1]$ for the diagonal self-attention weight at layer ℓ , head h , and position t , the executed score is

$$U_{\text{Attn}}(x; \hat{y}) := -\frac{1}{H} \sum_{h=1}^H \sum_{t=1}^T \log(\alpha_{\ell,h}(t, t) + \varepsilon),$$

where $\varepsilon = 10^{-12}$ is a numerical stabilizer and the layer ℓ is the model’s middle layer ($\ell = L/2$), as selected by LM-POLYGRAPH when no layer is explicitly specified. This choice matters: attention-based diagnostics vary substantially across layers, and the score should be understood as a layer-specific instantiation of the general idea [43].

RAUQ (Recurrent Attention Uncertainty Quantification). RAUQ propagates a confidence signal recursively along the sequence, combining the local token probability with attention from the current token to the previous one, and aggregates the per-layer scores only over the middle third of layers. Let $\mathcal{L} = \{\lfloor L/3 \rfloor, \dots, \lceil 2L/3 \rceil\}$ denote the indices of the middle third of the model’s L attention layers. For each $\ell \in \mathcal{L}$, the head attending most strongly (on average) to the previous token is

$$h^*(\ell) = \arg \max_h \frac{1}{T-1} \sum_{t=2}^T \alpha_{\ell,h}(t, t-1).$$

The recurrent confidence at layer ℓ is initialized at $C_1^{(\ell)} = p_\theta(\hat{y}_1 \mid x)$ and updated for $t > 1$ by

$$C_t^{(\ell)} = \alpha \cdot p_\theta(\hat{y}_t \mid x, \hat{y}_{<t}) + (1 - \alpha) \cdot \alpha_{\ell,h^*(\ell)}(t, t-1) \cdot C_{t-1}^{(\ell)},$$

where $\alpha \in [0, 1]$ is a mixing weight. The per-layer score is $1 - \frac{1}{T} \sum_t \log C_t^{(\ell)}$, and the estimator returns the worst (largest) layer over the middle third:

$$U_{\text{RAUQ}}(x; \hat{y}) := \max_{\ell \in \mathcal{L}} \left[1 - \frac{1}{T} \sum_{t=1}^T \log C_t^{(\ell)} \right].$$

Unlike AttentionScore, which evaluates each position independently, RAUQ captures how confidence accumulates or degrades along the sequence [50]. When α is not specified, LM-POLYGRAPH selects it through two binary flags: $\mathbf{1}_{\text{ent}} = 1$ if the entropy-based variant is used (and 0 for the probability variant), and $\mathbf{1}_{\text{instr}} = 1$ if the underlying model is instruction-tuned (and 0 otherwise). The default is

$$\alpha = (1 - \mathbf{1}_{\text{ent}}) (0.2 + 0.3 \cdot \mathbf{1}_{\text{instr}}) + \mathbf{1}_{\text{ent}} (0.8 + 0.1 \cdot \mathbf{1}_{\text{instr}}),$$

which yields $\alpha \in \{0.2, 0.5, 0.8, 0.9\}$ depending on the flags. We use the probability variant ($\mathbf{1}_{\text{ent}} = 0$) on instruction-tuned models ($\mathbf{1}_{\text{instr}} = 1$), so the executed $\alpha = 0.5$.

Contextualized Sequence Likelihood (CSL). Contextualized Sequence Likelihood reweights token negative log-likelihoods by an attention-derived saliency, so that positions the model deems important when producing the final response contribute more to the score than filler positions. Let $\bar{\alpha}_t = \text{mean}_{\ell,h} \alpha_{\ell,h}(\text{last}, t)$ be the mean attention paid to position t from the final generation step,

averaged over layers and heads, and let $w_t = \bar{\alpha}_t / \sum_{t'} \bar{\alpha}_{t'}$ be the corresponding saliency weight ($w_t \geq 0, \sum_t w_t = 1$). The executed score is

$$U_{\text{CSL}}(x; \hat{y}) := \sum_{t=1}^T w_t \cdot [-\log p_{\theta}(\hat{y}_t \mid x, \hat{y}_{<t})].$$

The score has the same form as TokenSAR but uses an internal saliency signal in place of an external cross-encoder relevance signal [32].

EigenScore. EigenScore measures the geometric spread of sampled internal embeddings in representation space. Let $\mathbf{e}^{(s)} \in \mathbb{R}^d$ denote the decoder hidden state of the last generated token for sample s , and let $\mathbf{E} = [\mathbf{e}^{(1)}, \dots, \mathbf{e}^{(S)}] \in \mathbb{R}^{d \times S}$ be the matrix whose columns are the S sample embeddings. Writing $\mathbf{J}_d = \mathbf{I}_d - \frac{1}{d} \mathbf{1}_d \mathbf{1}_d^\top \in \mathbb{R}^{d \times d}$ for the centering matrix and α for a regularization scalar, the regularized centered covariance matrix is

$$\mathbf{C} = \mathbf{E}^\top \mathbf{J}_d \mathbf{E} + \alpha \mathbf{I}_S \in \mathbb{R}^{S \times S},$$

with eigenvalues $\lambda_1, \dots, \lambda_S$. The executed score is the mean log-eigenvalue,

$$U_{\text{Eigen}}(x) := \frac{1}{S} \sum_{k=1}^S \log \lambda_k.$$

Sampled responses whose embeddings cluster in a low-dimensional subspace yield small eigenvalues and a low score (low uncertainty); sampled responses whose embeddings span many directions yield large eigenvalues and a high score (high uncertainty) [7]. We use $\alpha = 10^{-3}$.

B.5.4 Training-based estimators

Training-based estimators fit a reference distribution in representation space on a held-out training partition and score test inputs by their distance from that reference. They are best read as familiarity or out-of-distribution signals rather than as readouts of response confidence. All variants below operate on decoder hidden states averaged over the generated tokens of $\hat{y}$, and are all fitted to each task’s own training split. The relative variants additionally subtract a background distance term fitted on a broad, general-purpose corpus that we describe below.

Mahalanobis Distance. Mahalanobis Distance is the basic representation-space estimator. Let $\mathbf{e}(x, \hat{y}) \in \mathbb{R}^d$ denote the decoder hidden state averaged over all generated tokens of $\hat{y}$, and let $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ be the mean and covariance estimated from the task’s training partition. The executed score is the Mahalanobis distance between $\mathbf{e}(x, \hat{y})$ and the training centroid:

$$U_{\text{MD}}(x; \hat{y}) := \sqrt{(\mathbf{e}(x, \hat{y}) - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{e}(x, \hat{y}) - \boldsymbol{\mu})}.$$

A test input whose representation lies near the training centroid receives a small distance (low uncertainty); an input far from the training distribution receives a large distance (high uncertainty) [29].

Relative Mahalanobis Distance. Relative Mahalanobis Distance subtracts a background Mahalanobis distance, fitted on a broad general-purpose corpus, in order to isolate task-specific novelty from generic rarity:

$$U_{\text{RMD}}(x; \hat{y}) := U_{\text{MD}}(x; \hat{y}) - U_{\text{MD}_0}(x; \hat{y}),$$

where U_{MD_0} uses the background mean and covariance in place of the task mean and covariance. A test input far from both the task and the background receives a small score (general rarity); an input far from the task but close to the background receives a large score, indicating domain shift specific to the task [40]. We fit the background distribution on allenai/c4 (shard en/c4-train.00000-of-01024.json.gz).⁶

⁶<https://huggingface.co/datasets/allenai/c4>

Robust Density Estimation (RDE). Robust Density Estimation replaces the Gaussian fit of Mahalanobis Distance with a robust covariance estimate computed over a Kernel-PCA-projected representation space, to better handle non-elliptical embedding distributions and reduce sensitivity to training-set outliers. Let $\tilde{\mathbf{e}}(x; \hat{y}) \in \mathbb{R}^{d'}$ denote the Kernel-PCA projection of $\mathbf{e}(x, \hat{y})$, and let $\hat{\boldsymbol{\mu}} \in \mathbb{R}^{d'}$ and $\hat{\boldsymbol{\Sigma}} \in \mathbb{R}^{d' \times d'}$ be the mean and covariance estimated by the Minimum Covariance Determinant procedure in the projected space. The executed score is the Mahalanobis distance in this robust, dimensionality-reduced space:

$$U_{\text{RDE}}(x; \hat{y}) := \sqrt{(\tilde{\mathbf{e}}(x, \hat{y}) - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\tilde{\mathbf{e}}(x, \hat{y}) - \hat{\boldsymbol{\mu}})}.$$

[57].

HUQ-MD and HUQ-RMD. The HUQ variants combine a representation-space distance (epistemic-novelty signal) with a probability-based uncertainty (aleatoric signal) via *empirical-CDF rank fusion* [49]. Because Perplexity and Mahalanobis Distance have different scales and distributions, one cannot combine them by raw averaging; rank fusion instead converts each score to a scale-free percentile relative to the training partition before averaging. Concretely, let $\mathcal{D}_{\text{train}}$ denote the task’s training partition. The empirical CDF of U_{PPL} on that partition is

$$F_{\text{PPL}}^{\text{train}}(u) = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{x' \in \mathcal{D}_{\text{train}}} \mathbf{1}[U_{\text{PPL}}(x', \hat{y}(x')) \leq u] \in [0, 1],$$

and $F_{\text{MD}}^{\text{train}}$ is defined analogously for U_{MD} . Evaluating these CDFs at a test input gives its percentile rank on each score (the fraction of training inputs with a lower score), and the executed HUQ-MD score is the mean of the two percentile ranks:

$$U_{\text{HUQ-MD}}(x; \hat{y}) := \frac{1}{2} [F_{\text{PPL}}^{\text{train}}(U_{\text{PPL}}(x; \hat{y})) + F_{\text{MD}}^{\text{train}}(U_{\text{MD}}(x; \hat{y}))] \in [0, 1].$$

HUQ-RMD is defined analogously with U_{RMD} in place of U_{MD} . Because both components are converted to $[0, 1]$ -valued percentile ranks before averaging, the combination is scale-free and distribution-free.

B.5.5 Reflexive estimators (white-box)

Reflexive estimators prompt the model to judge its own answer. The white-box variants additionally require direct access to the probability assigned to the “True” option; their black-box counterpart is described in Section B.6.4.

PTrue. Given an input x and a proposed answer $\hat{y}$, PTrue constructs a self-evaluation prompt $\text{prompt}_{\text{PT}}(x; \hat{y})$ asking the model to judge whether $\hat{y}$ is correct and reads off the probability assigned to the “True” option. The executed score is

$$U_{\text{PTrue}}(x; \hat{y}) := -\log p_{\theta}(\text{“True”} \mid \text{prompt}_{\text{PT}}(x; \hat{y})).$$

A high probability of “True” yields a small score (low uncertainty); a low probability yields a large score (high uncertainty) [25]. The exact prompt template is:

Question: {x}.
Proposed Answer: {y}.
Is the proposed answer True or False?

PTrue Sampling. PTrue Sampling augments the self-evaluation prompt with S sampled candidate answers $y^{(1)}, \dots, y^{(S)}$ before asking the truth judgment, in order to expose the model to its own response variability:

$$U_{\text{PTrue-S}}(x; \hat{y}) := -\log p_{\theta}(\text{“True”} \mid \text{prompt}_{\text{PT-S}}(x, \hat{y}, y^{(1)}, \dots, y^{(S)})).$$

The exact prompt template is:

Question: {x}
Here are some ideas that were brainstormed: {y⁽¹⁾, ..., y^(S)}
Possible answer: {y}
Is the possible answer:
(A) True
(B) False
The possible answer is:

B.6 Black-box estimators

Black-box estimators operate only on sampled responses and external semantic/lexical-relation functions, without using token probabilities, attention, or hidden states. We organize them into four groups: the simplest estimators that summarize the semantic-class structure of the samples (NumSet, LabelProb); spectral and graph-density estimators on a pairwise similarity graph (Kernel Language Entropy, EigValLaplacian, Eccentricity, DegMat, LUQ); lexical baselines that replace semantic relations with pairwise surface overlap (ROUGE-L Similarity, BLEU Similarity); and a reflexive black-box estimator (PTrue Empirical).

B.6.1 Semantic-class summaries

NumSet. NumSet is the simplest semantic-consistency estimator: it counts the number of distinct semantic classes among the sampled responses. Samples are partitioned into bidirectional-entailment classes $C_1, \dots, C_M$, as in Semantic Entropy, and the executed score is the cluster count:

$$U_{\text{NumSet}}(x) := M.$$

The score is integer-valued, with a minimum of 1 when all samples express the same meaning and a maximum of S when every sample expresses a different meaning [33]. The granularity is coarse, but the score is robust to NLI-model calibration.

LabelProb. We retain the package name LABELPROB for consistency with LM-POLYGRAPH, but the executed implementation is more accurately described as the complement of the dominant semantic-class frequency:

$$U_{\text{LabelProb}}(x) := 1 - \max_{1 \leq m \leq M} \frac{|C_m|}{S}.$$

The score is zero when a single semantic class accounts for all samples, and approaches $1 - 1/S$ when every sample belongs to a different class. LabelProb is therefore best understood as a majority-class complement, not as a black-box approximation of maximum sequence probability as the name might suggest [47].

B.6.2 Spectral and graph-density estimators

These estimators build a pairwise relation matrix $\mathbf{W} = (W_{ij})_{i,j=1}^S \in \mathbb{R}^{S \times S}$ over sampled responses and summarize its topology. Three of them rely on spectral analysis of the corresponding graph (Kernel Language Entropy uses kernel eigenvalues; EigValLaplacian uses Laplacian eigenvalues; Eccentricity uses Laplacian eigenvectors), and the remaining two summarize the graph through its degree structure (DegMat) or through average pairwise scores (LUQ). For EigValLaplacian, DegMat, and Eccentricity, we run three edge constructions: NLI-entailment ($W_{ij} = E_{ij}$), NLI-contradiction ($W_{ij} = 1 - C_{ij}$), and Jaccard token similarity.

Kernel Language Entropy (KLE). KLE quantifies semantic uncertainty as the von Neumann entropy of a heat-diffusion kernel on the semantic graph. Let $\mathbf{D} = \text{diag}(\sum_j W_{ij})$ be the degree matrix, $\mathbf{L} = \mathbf{D} - \mathbf{W}$ the unnormalized graph Laplacian, and $\mathbf{K} = \exp(-t\mathbf{L})$ the heat kernel (subsequently row-normalized to unit trace). Writing $\lambda_k(\mathbf{K})$ for the eigenvalues of $\mathbf{K}$:

$$U_{\text{KLE}}(x) := -\text{tr}(\mathbf{K} \log \mathbf{K}) = -\sum_k \lambda_k(\mathbf{K}) \log \lambda_k(\mathbf{K}).$$

A graph with many disconnected semantic clusters yields several large eigenvalues and a high entropy; a fully connected graph yields a single dominant eigenvalue and low entropy [36].

EigValLaplacian. EigValLaplacian also operates on the spectrum of the graph, but on the eigenvalues of the normalized Laplacian $\tilde{\mathbf{L}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}$ rather than on a heat kernel. Writing $\lambda_1, \dots, \lambda_S \in [0, 2]$ for its eigenvalues:

$$U_{\text{EVL}}(x) := \sum_{k=1}^S \max(0, 1 - \lambda_k).$$

Each term measures how close one mode of variation is to being fully disconnected; the sum is a continuous relaxation of the number of semantic sets [33].

Eccentricity. Eccentricity (Ecc) uses the *eigenvectors* of $\tilde{\mathbf{L}}$ rather than its eigenvalues. Let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be the eigenvectors of $\tilde{\mathbf{L}}$ corresponding to its smallest k eigenvalues, and stack their per-sample entries into spectral embeddings $\mathbf{v}_i = (\mathbf{u}_1[i], \dots, \mathbf{u}_k[i]) \in \mathbb{R}^k$ for sample i . Writing $\bar{\mathbf{v}} = \frac{1}{S} \sum_i \mathbf{v}_i$ for the centroid:

$$U_{\text{Ecc}}(x) := \left(\sum_{i=1}^S \|\mathbf{v}_i - \bar{\mathbf{v}}\|_2^2 \right)^{1/2}.$$

A graph in which all samples occupy a tight region of the spectral embedding space yields a small score; a graph with spectral outliers yields a large score [33].

DegMat. Degree Matrix (DegMat) summarizes a graph’s degree structure rather than its spectrum. The executed score is the mean missing pairwise similarity:

$$U_{\text{DegMat}}(x) := \frac{S^2 - \sum_{i=1}^S \sum_{j=1}^S W_{ij}}{S^2},$$

which equals zero when all pairs have $W_{ij} = 1$ (full mutual agreement) and approaches 1 when no pairs agree. Like Eccentricity, DegMat depends on the choice of edge construction, and we report all three variants [33].

LUQ. Long-text Uncertainty Quantification (LUQ) summarizes the graph through average pairwise NLI confidence, using the continuous softmax score from the NLI logits rather than discretized entail/contradict decisions. Writing $\sigma_{ij} = \exp(z_{ij}^E) / (\exp(z_{ij}^E) + \exp(z_{ij}^C))$ for the soft entailment confidence between $y^{(i)}$ and $y^{(j)}$:

$$U_{\text{LUQ}}(x) := 1 - \frac{1}{S(S-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^S \sigma_{ij}.$$

A high average pairwise entailment confidence yields a small score (low uncertainty); weak average confidence yields a large score [59].

B.6.3 Lexical similarity baselines

ROUGE-L Similarity and BLEU Similarity. These baselines replace NLI-based semantic relations with pairwise surface-overlap measures. For $\text{metric}(\cdot, \cdot) \in \{\text{ROUGE-L}, \text{BLEU}\}$, the executed score is the negated mean pairwise similarity:

$$U_{\text{LexSim}}(x) := -\frac{2}{S(S-1)} \sum_{1 \leq i < j \leq S} \text{metric}(y^{(i)}, y^{(j)}).$$

Surface-form variation across samples yields a high score even when the samples convey the same meaning, so these baselines cannot distinguish valid paraphrasing from semantic disagreement [30, 38]. We report ROUGE-L and BLEU variants separately.

B.6.4 Reflexive estimator (black-box)

PTrue Empirical. PTrue Empirical is the black-box counterpart of PTrue: instead of reading off the probability of the “True” token, it draws N stochastic self-evaluation responses from the model and measures the empirical frequency with which the model judges its own answer to be true:

$$U_{\text{PTrue-E}}(x; \hat{y}) := 1 - \frac{1}{N} \sum_{n=1}^N \mathbf{1}[\text{“True”} \in \text{sample}_n(\text{prompt}_{\text{PT}}(x; \hat{y}))],$$

using the same self-evaluation prompt as PTrue. We use $N = 10$ [44].

Estimator	Samples	Hyperparameters and external models
<i>logit-based (Sec. B.5.1)</i>		
Rényi Divergence	1	$\alpha = 0.5, \tau = 2$
Fisher–Rao Distance	1	$\tau = 2$
CPMI	1	$\tau = 0.0656, \lambda = 3.599$
TokenSAR	1	cross-encoder stsb-roberta-large
CCP	1	NLI deberta-large-mnli; top- k alternatives, $k = 10$
<i>Sample semantic dispersion (Sec. B.5.2)</i>		
MC-SE, MC-NSE	$S = 10$	sampling temperature 1
Semantic Entropy	$S = 10$	NLI deberta-large-mnli
Semantic Density	$S = 10$	NLI deberta-large-mnli
SentenceSAR	$S = 10$	cross-encoder stsb-roberta-large; $\tau = 1$
SAR	$S = 10$	cross-encoder stsb-roberta-large; $\tau = 1$
Cocoa{MSP, PPL, MTE}	$S = 10$	cross-encoder stsb-roberta-large;
<i>Internal state (Sec. B.5.3)</i>		
AttentionScore	1	layer $\ell = L/2; \varepsilon = 10^{-12}$
RAUQ	1	$\alpha = 0.5$; middle-third layers $\mathcal{L} = \{\lfloor L/3 \rfloor, \dots, \lceil 2L/3 \rceil\}$
EigenScore	$S = 10$	decoder hidden state of last token; $\alpha = 10^{-3}$
<i>Training-based (Sec. B.5.4)</i>		
Mahalanobis Distance	1	decoder hidden states; fitted per task on training partition
Relative MD	1	background: allenai/c4, shard en/c4-train.00000
RDE	1	Kernel-PCA projection + MCD covariance
HUQ-MD, HUQ-RMD	1	CDF rank fusion of U_{PPL} and $U_{\text{MD}} / U_{\text{RMD}}$ on training partition
<i>Reflexive, white-box (Sec. B.5.5)</i>		
PTrue	1	prompt: see §B.5.5
PTrue Sampling	$S = 10$	prompt: see §B.5.5
<i>Black-box (Sec. B.6)</i>		
NumSet, LabelProb	$S = 10$	NLI deberta-large-mnli
KLE	$S = 10$	NLI deberta-large-mnli; heat-kernel time $t = 0.3$
EigValLaplacian, Ecc, DegMat	$S = 10$	NLI deberta-large-mnli;
LUQ	$S = 10$	NLI deberta-large-mnli
Lexical Similarity	$S = 10$	metrics: ROUGE-L, BLEU
PTrue Empirical	$N = 10$ self-eval.	prompt: see §B.6.4

Table 2: Numerical hyperparameters and external models for every estimator executed in our benchmark. “Samples” is the number of stochastic generations generated per input. Estimators that have no tunable hyperparameters and no external models (MSP, Perplexity, Mean Token Entropy, SelfCertainty, PMI, MC-SE, MC-NSE, CSL) are omitted from the table. Across all sample-based estimators, the sample pool is shared, the sampling temperature is 1, and the random seed is 42.

C Methodology details

C.1 Asbtention Template

Abstention template. “No information is available about this, and I am not aware of any such entity being recognized or documented. This name does not appear to correspond to anything known, and I do not believe this exists as a real or established thing. There may be some confusion, or this could be a misspelling or an alternative rendering of a different name. I am not certain what this refers to, and this term is not widely recognized in any domain I can identify. If you could provide more context or clarify your intent, that would help determine whether a related concept exists.”

D Complementary experimental results

Table 3: AUROC $\uparrow$, PRR $\uparrow$, RCE $\downarrow$ for **All Models** across tasks (fixed metric-wise gradient; warmer = higher value). Uncertainty estimators and AlignScore variants are ranked separately: bold = best per column (ties marked); underline = second-best (ties marked) within the uncertainty group; AlignScore group shows only the best.

Uncertainty Metric	AUROC $\uparrow$	RAGTruth PRR $\uparrow$	RCE $\downarrow$	AUROC $\uparrow$	PreciseWiki PRR $\uparrow$	RCE $\downarrow$	AUROC $\uparrow$	LongWiki PRR $\uparrow$	RCE $\downarrow$	AUROC $\uparrow$	NonExistent PRR $\uparrow$	RCE $\downarrow$
Information-based												
Maximum Seq. Probability (MSP)	<u>0.64$\pm$0.10</u>	<u>0.28$\pm$0.26</u>	<u>0.33$\pm$0.24</u>	0.71 $\pm$ 0.05	<u>0.56$\pm$0.08</u>	0.28 $\pm$ 0.07	<u>0.56$\pm$0.03</u>	0.21 $\pm$ 0.06	<u>0.44$\pm$0.13</u>	<u>0.67$\pm$0.08</u>	<u>0.38$\pm$0.19</u>	<u>0.38$\pm$0.18</u>
Perplexity (PPL)	0.44 $\pm$ 0.09	-0.23 $\pm$ 0.17	<u>0.83$\pm$0.24</u>	0.65 $\pm$ 0.07	0.45 $\pm$ 0.26	0.33 $\pm$ 0.20	0.57 $\pm$ 0.03	0.23 $\pm$ 0.07	0.37 $\pm$ 0.15	0.43 $\pm$ 0.13	-0.32 $\pm$ 0.49	0.74 $\pm$ 0.21
Mean Token Entropy (MTE)	0.52 $\pm$ 0.01	-0.08 $\pm$ 0.05	0.65 $\pm$ 0.07	0.64 $\pm$ 0.07	0.46 $\pm$ 0.22	0.35 $\pm$ 0.17	0.57 $\pm$ 0.03	0.21 $\pm$ 0.09	0.40 $\pm$ 0.15	0.44 $\pm$ 0.13	-0.26 $\pm$ 0.48	0.74 $\pm$ 0.20
Pointwise Mutual Info. (PMI)	0.33 $\pm$ 0.01	-0.50 $\pm$ 0.06	<u>0.99$\pm$0.01</u>	0.47 $\pm$ 0.00	-0.17 $\pm$ 0.05	<u>0.89$\pm$0.03</u>	0.51 $\pm$ 0.02	0.00 $\pm$ 0.09	0.68 $\pm$ 0.12	0.44 $\pm$ 0.02	-0.11 $\pm$ 0.09	0.78 $\pm$ 0.06
Conditional PMI	0.51 $\pm$ 0.07	-0.09 $\pm$ 0.20	0.69 $\pm$ 0.20	0.31 $\pm$ 0.03	<u>0.75$\pm$0.14</u>	<u>0.59$\pm$0.08</u>	0.43 $\pm$ 0.03	-0.22 $\pm$ 0.11	<u>0.64$\pm$0.07</u>	0.57 $\pm$ 0.13	0.01 $\pm$ 0.36	0.61 $\pm$ 0.31
Self Certainty	0.46 $\pm$ 0.02	-0.23 $\pm$ 0.05	0.82 $\pm$ 0.10	0.57 $\pm$ 0.02	0.34 $\pm$ 0.14	0.48 $\pm$ 0.05	0.55 $\pm$ 0.02	0.13 $\pm$ 0.06	0.47 $\pm$ 0.14	0.50 $\pm$ 0.08	-0.07 $\pm$ 0.28	0.62 $\pm$ 0.12
Rényi Divergence	0.45 $\pm$ 0.02	-0.26 $\pm$ 0.04	<u>0.89$\pm$0.08</u>	0.55 $\pm$ 0.02	0.28 $\pm$ 0.12	0.54 $\pm$ 0.05	0.54 $\pm$ 0.02	0.10 $\pm$ 0.06	0.49 $\pm$ 0.13	0.50 $\pm$ 0.07	-0.09 $\pm$ 0.27	0.59 $\pm$ 0.11
Fisher-Rao Distance	0.47 $\pm$ 0.03	-0.22 $\pm$ 0.06	0.81 $\pm$ 0.11	0.55 $\pm$ 0.03	0.26 $\pm$ 0.15	0.55 $\pm$ 0.09	0.55 $\pm$ 0.02	0.15 $\pm$ 0.07	0.46 $\pm$ 0.15	0.53 $\pm$ 0.07	-0.02 $\pm$ 0.24	0.54 $\pm$ 0.06
Token SAR	0.44 $\pm$ 0.09	-0.23 $\pm$ 0.17	<u>0.83$\pm$0.24</u>	0.71 $\pm$ 0.05	<u>0.57$\pm$0.19</u>	0.21 $\pm$ 0.10	0.57 $\pm$ 0.03	0.23 $\pm$ 0.07	0.36 $\pm$ 0.15	0.42 $\pm$ 0.15	-0.39 $\pm$ 0.58	0.74 $\pm$ 0.20
Chain-Conditioned Probability (CCP)	0.61 $\pm$ 0.11	0.23 $\pm$ 0.25	0.40 $\pm$ 0.30	0.72 $\pm$ 0.02	0.61 $\pm$ 0.01	0.21 $\pm$ 0.04	0.62 $\pm$ 0.02	0.40 $\pm$ 0.05	0.28 $\pm$ 0.08	0.73$\pm$0.09	0.55$\pm$0.18	0.30$\pm$0.11
Sampling-based												
Monte-Carlo Seq. Entropy	0.42 $\pm$ 0.06	-0.36 $\pm$ 0.14	<u>0.87$\pm$0.09</u>	0.67 $\pm$ 0.07	0.48 $\pm$ 0.18	0.27 $\pm$ 0.11	0.52 $\pm$ 0.01	0.05 $\pm$ 0.05	0.61 $\pm$ 0.06	0.58 $\pm$ 0.06	0.23 $\pm$ 0.15	0.52 $\pm$ 0.14
Monte-Carlo Norm. Seq. Entropy	0.37 $\pm$ 0.02	-0.43 $\pm$ 0.10	<u>1.00$\pm$0.00</u>	0.67 $\pm$ 0.06	0.49 $\pm$ 0.18	0.27 $\pm$ 0.14	0.53 $\pm$ 0.01	0.06 $\pm$ 0.05	0.58 $\pm$ 0.09	0.48 $\pm$ 0.08	-0.09 $\pm$ 0.27	0.78 $\pm$ 0.24
Semantic Entropy	0.44 $\pm$ 0.06	-0.31 $\pm$ 0.15	<u>0.87$\pm$0.09</u>	0.69 $\pm$ 0.07	0.59 $\pm$ 0.16	0.29 $\pm$ 0.07	0.54 $\pm$ 0.03	0.12 $\pm$ 0.09	0.54 $\pm$ 0.10	0.58 $\pm$ 0.06	0.21 $\pm$ 0.16	0.55 $\pm$ 0.15
Semantic Density	0.45 $\pm$ 0.05	-0.18 $\pm$ 0.17	0.87 $\pm$ 0.08	0.75 $\pm$ 0.02	0.61 $\pm$ 0.10	0.24 $\pm$ 0.15	0.59 $\pm$ 0.02	0.29 $\pm$ 0.09	0.32 $\pm$ 0.07	0.42 $\pm$ 0.07	-0.22 $\pm$ 0.22	<u>0.80$\pm$0.09</u>
Sentence SAR	0.51 $\pm$ 0.08	-0.14 $\pm$ 0.21	0.68 $\pm$ 0.19	0.72 $\pm$ 0.06	0.59 $\pm$ 0.10	0.27 $\pm$ 0.06	0.51 $\pm$ 0.02	0.05 $\pm$ 0.06	0.67 $\pm$ 0.11	0.66 $\pm$ 0.05	0.47 $\pm$ 0.09	0.31 $\pm$ 0.02
SAR	0.41 $\pm$ 0.05	-0.32 $\pm$ 0.12	<u>0.93$\pm$0.08</u>	0.72 $\pm$ 0.05	0.58 $\pm$ 0.12	0.24 $\pm$ 0.05	0.59 $\pm$ 0.02	0.23 $\pm$ 0.04	0.38 $\pm$ 0.06	0.43 $\pm$ 0.08	-0.23 $\pm$ 0.27	<u>0.85$\pm$0.17</u>
Cocoa MSP	0.62 $\pm$ 0.10	0.24 $\pm$ 0.25	0.39 $\pm$ 0.28	0.74 $\pm$ 0.04	0.63 $\pm$ 0.04	0.23 $\pm$ 0.02	0.61 $\pm$ 0.03	0.33 $\pm$ 0.09	0.30 $\pm$ 0.14	0.60 $\pm$ 0.08	0.22 $\pm$ 0.25	0.47 $\pm$ 0.23
Cocoa PPL	0.45 $\pm$ 0.07	-0.21 $\pm$ 0.16	<u>0.84$\pm$0.22</u>	0.69 $\pm$ 0.06	0.55 $\pm$ 0.17	0.26 $\pm$ 0.12	0.62 $\pm$ 0.04	0.35 $\pm$ 0.09	0.25 $\pm$ 0.10	0.39 $\pm$ 0.12	-0.45 $\pm$ 0.47	0.87 $\pm$ 0.11
Cocoa MTE	0.51 $\pm$ 0.02	-0.10 $\pm$ 0.06	0.69 $\pm$ 0.10	0.69 $\pm$ 0.06	0.57 $\pm$ 0.15	0.23 $\pm$ 0.08	0.62 $\pm$ 0.04	0.34 $\pm$ 0.09	0.24 $\pm$ 0.11	0.40 $\pm$ 0.12	-0.39 $\pm$ 0.46	0.85 $\pm$ 0.13
Internal												
Attention Score	0.69$\pm$0.03	0.41$\pm$0.09	0.28$\pm$0.03	0.59 $\pm$ 0.03	0.31 $\pm$ 0.03	0.41 $\pm$ 0.10	0.47 $\pm$ 0.01	-0.12 $\pm$ 0.04	<u>0.87$\pm$0.08</u>	0.72 $\pm$ 0.11	0.48 $\pm$ 0.20	0.42 $\pm$ 0.06
RAUQ	0.44 $\pm$ 0.09	-0.22 $\pm$ 0.17	<u>0.82$\pm$0.24</u>	0.68 $\pm$ 0.09	0.48 $\pm$ 0.24	0.34 $\pm$ 0.23	0.56 $\pm$ 0.03	0.23 $\pm$ 0.06	0.41 $\pm$ 0.15	0.44 $\pm$ 0.12	-0.28 $\pm$ 0.47	0.74 $\pm$ 0.20
CSL	0.39 $\pm$ 0.02	-0.37 $\pm$ 0.03	<u>0.97$\pm$0.04</u>	0.63 $\pm$ 0.04	0.40 $\pm$ 0.09	0.35 $\pm$ 0.10	0.53 $\pm$ 0.03	0.10 $\pm$ 0.09	0.54 $\pm$ 0.17	0.47 $\pm$ 0.01	-0.15 $\pm$ 0.14	<u>0.83$\pm$0.03</u>
EigenScore	0.44 $\pm$ 0.08	-0.29 $\pm$ 0.22	<u>0.87$\pm$0.09</u>	0.62 $\pm$ 0.06	0.23 $\pm$ 0.15	0.50 $\pm$ 0.09	0.47 $\pm$ 0.03	-0.10 $\pm$ 0.13	<u>0.84$\pm$0.11</u>	0.56 $\pm$ 0.10	0.11 $\pm$ 0.24	0.66 $\pm$ 0.24
Training-based												
Mahalanobis Distance (MD)	0.27 $\pm$ 0.04	-0.48 $\pm$ 0.15	<u>1.00$\pm$0.00</u>	0.50 $\pm$ 0.02	0.03 $\pm$ 0.03	0.68 $\pm$ 0.07	0.40 $\pm$ 0.04	-0.32 $\pm$ 0.16	<u>0.97$\pm$0.04</u>	0.35 $\pm$ 0.18	-0.77 $\pm$ 0.71	0.81 $\pm$ 0.26
Relative MD (RMD)	0.29 $\pm$ 0.03	-0.43 $\pm$ 0.08	<u>1.00$\pm$0.00</u>	0.49 $\pm$ 0.02	0.00 $\pm$ 0.04	0.74 $\pm$ 0.07	0.42 $\pm$ 0.05	-0.25 $\pm$ 0.19	<u>0.93$\pm$0.08</u>	0.30 $\pm$ 0.15	-0.91 $\pm$ 0.69	0.88 $\pm$ 0.17
Robust Density Estimation (RDE)	0.32 $\pm$ 0.05	-0.23 $\pm$ 0.14	<u>0.98$\pm$0.01</u>	0.51 $\pm$ 0.09	0.02 $\pm$ 0.04	0.62 $\pm$ 0.28	0.49 $\pm$ 0.06	-0.02 $\pm$ 0.20	<u>0.76$\pm$0.28</u>	0.36 $\pm$ 0.12	-0.50 $\pm$ 0.51	0.90 $\pm$ 0.15
HUQ-MD	0.44 $\pm$ 0.09	-0.23 $\pm$ 0.17	0.83 $\pm$ 0.24	0.64 $\pm$ 0.07	0.45 $\pm$ 0.27	0.32 $\pm$ 0.21	0.57 $\pm$ 0.03	0.23 $\pm$ 0.07	0.37 $\pm$ 0.15	0.43 $\pm$ 0.13	-0.32 $\pm$ 0.49	0.74 $\pm$ 0.21
HUQ-RMD	0.44 $\pm$ 0.08	-0.22 $\pm$ 0.17	<u>0.83$\pm$0.24</u>	0.64 $\pm$ 0.07	0.45 $\pm$ 0.27	0.32 $\pm$ 0.21	0.57 $\pm$ 0.03	0.23 $\pm$ 0.07	0.37 $\pm$ 0.15	0.43 $\pm$ 0.13	-0.32 $\pm$ 0.49	0.74 $\pm$ 0.21
Reflexive												
PTrue	0.51 $\pm$ 0.09	-0.04 $\pm$ 0.21	0.63 $\pm$ 0.33	0.51 $\pm$ 0.05	0.10 $\pm$ 0.15	0.70 $\pm$ 0.19	0.47 $\pm$ 0.01	-0.11 $\pm$ 0.04	<u>0.87$\pm$0.06</u>	0.42 $\pm$ 0.15	-0.43 $\pm$ 0.67	0.76 $\pm$ 0.18
PTrue Sampling	0.59 $\pm$ 0.10	-0.03 $\pm$ 0.25	0.64 $\pm$ 0.33	0.51 $\pm$ 0.03	0.04 $\pm$ 0.15	0.63 $\pm$ 0.12	0.49 $\pm$ 0.00	-0.04 $\pm$ 0.03	<u>0.80$\pm$0.04</u>	0.45 $\pm$ 0.09	-0.17 $\pm$ 0.30	0.79 $\pm$ 0.16
PTrue Empirical	0.57 $\pm$ 0.08	0.16 $\pm$ 0.17	0.45 $\pm$ 0.24	0.49 $\pm$ 0.03	-0.02 $\pm$ 0.06	0.72 $\pm$ 0.11	0.49 $\pm$ 0.00	-0.04 $\pm$ 0.03	<u>0.80$\pm$0.04</u>	0.45 $\pm$ 0.09	-0.17 $\pm$ 0.30	0.79 $\pm$ 0.16
Black-box												
NumSet	0.47 $\pm$ 0.01	-0.09 $\pm$ 0.01	0.85 $\pm$ 0.05	0.74 $\pm$ 0.03	0.64 $\pm$ 0.08	0.16$\pm$0.05	0.53 $\pm$ 0.01	0.13 $\pm$ 0.06	0.58 $\pm$ 0.03	0.45 $\pm$ 0.01	-0.24 $\pm$ 0.03	<u>0.91$\pm$0.05</u>
LabelProb	0.46 $\pm$ 0.08	-0.15 $\pm$ 0.03	0.74 $\pm$ 0.19	0.71 $\pm$ 0.04	0.52 $\pm$ 0.10	0.25 $\pm$ 0.08	0.54 $\pm$ 0.03	0.12 $\pm$ 0.10	0.51 $\pm$ 0.14	0.51 $\pm$ 0.06	0.01 $\pm$ 0.15	0.72 $\pm$ 0.22
Kernel Language Entropy (KLE)	0.43 $\pm$ 0.07	-0.23 $\pm$ 0.19	0.85 $\pm$ 0.12	0.74 $\pm$ 0.04	0.61 $\pm$ 0.12	0.24 $\pm$ 0.13	0.60 $\pm$ 0.03	0.30 $\pm$ 0.09	0.30 $\pm$ 0.07	0.47 $\pm$ 0.05	-0.14 $\pm$ 0.10	0.82 $\pm$ 0.12
EigValLap NLI (Entail)	0.44 $\pm$ 0.08	-0.20 $\pm$ 0.22	0.83 $\pm$ 0.14	0.75 $\pm$ 0.04	0.64 $\pm$ 0.10	0.21 $\pm$ 0.09	0.59 $\pm$ 0.03	0.24 $\pm$ 0.08	0.34 $\pm$ 0.08	0.47 $\pm$ 0.05	-0.09 $\pm$ 0.17	0.81 $\pm$ 0.14
EigValLap NLI (Contra)	0.42 $\pm$ 0.02	-0.17 $\pm$ 0.06	<u>0.98$\pm$0.02</u>	0.67 $\pm$ 0.07	0.44 $\pm$ 0.20	0.32 $\pm$ 0.17	0.65$\pm$0.03	0.41$\pm$0.06	<u>0.22$\pm$0.09</u>	0.48 $\pm$ 0.04	-0.05 $\pm$ 0.11	0.83 $\pm$ 0.13
EigValLap Jaccard	0.36 $\pm$ 0.05	-0.43 $\pm$ 0.16	<u>0.98$\pm$0.03</u>	0.65 $\pm$ 0.02	0.43 $\pm$ 0.08	0.29 $\pm$ 0.08	0.58 $\pm$ 0.02	0.25 $\pm$ 0.05	0.36 $\pm$ 0.09	0.39 $\pm$ 0.05	-0.43 $\pm$ 0.19	<u>0.97$\pm$0.04</u>
Eccentricity NLI (Entail)	0.45 $\pm$ 0.09	-0.16 $\pm$ 0.25	0.70 $\pm$ 0.18	0.72 $\pm$ 0.04	0.52 $\pm$ 0.10	0.30 $\pm$ 0.04	0.58 $\pm$ 0.03	0.20 $\pm$ 0.10	0.40 $\pm$ 0.09	0.50 $\pm$ 0.05	0.01 $\pm$ 0.11	0.74 $\pm$ 0.16
Eccentricity NLI (Contra)	0.41 $\pm$ 0.01	-0.17 $\pm$ 0.06	<u>0.98$\pm$0.02</u>	0.73 $\pm$ 0.04	0.62 $\pm$ 0.09	0.20 $\pm$ 0.07	0.64 $\pm$ 0.02	0.37 $\pm$ 0.05	0.23 $\pm$ 0.05	0.48 $\pm$ 0.05	-0.06 $\pm$ 0.16	0.83 $\pm$ 0.16
Eccentricity Jaccard	0.54 $\pm$ 0.04	0.05 $\pm$ 0.06	0.63 $\pm$ 0.16	0.66 $\pm$ 0.04	0.41 $\pm$ 0.11	0.30 $\pm$ 0.08	0.49 $\pm$ 0.01	-0.02 $\pm$ 0.03	<u>0.78$\pm$0.06</u>	0.52 $\pm$ 0.01	0.08 $\pm$ 0.06	0.62 $\pm$ 0.07
DegMat NLI (Entail)	0.44 $\pm$ 0.08	-0.21 $\pm$ 0.22	0.80 $\pm$ 0.15	0.75$\pm$0.03	0.65$\pm$0.08	0.21 $\pm$ 0.11	0.57 $\pm$ 0.03	0.21 $\pm$ 0.09	0.38 $\pm$ 0.10	0.47 $\pm$ 0.06	-0.07 $\pm$ 0.20	0.79 $\pm$ 0.17
DegMat NLI (Contra)	0.40 $\pm$ 0.02	-0.23 $\pm$ 0.07	<u>0.99$\pm$0.01</u>	0.68 $\pm$ 0.06	0.47 $\pm$ 0.18	0.30 $\pm$ 0.15	0.65 $\pm$ 0.03	0.40 $\pm$ 0.08	0.22$\pm$0.07	0.48 $\pm$ 0.04	-0.09 $\pm$ 0.10	0.83 $\pm$ 0.13
DegMat Jaccard	0.38 $\pm$ 0.06	-0.40 $\pm$ 0.19	0.94 $\pm$ 0.06	0.66 $\pm$ 0.03	0.44 $\pm$ 0.07	0.27 $\pm$ 0.08	0.59 $\pm$ 0.02	0.26 $\pm$ 0.06	0.34 $\pm$ 0.11	0.39 $\pm$ 0.05	-0.43 $\pm$ 0.19	0.96 $\pm$ 0.05
LUQ	0.38 $\pm$ 0.03	-0.29 $\pm$ 0.11	0.98 $\pm$ 0.03	0.74 $\pm$ 0.04	0.62 $\pm$ 0.11	0.22 $\pm$ 0.11	<u>0.65$\pm$0.03</u>	0.39 $\pm$ 0.08	0.24 $\pm$ 0.03	0.45 $\pm$ 0.04	-0.21 $\pm$ 0.08	0.85 $\pm$ 0.10
Lexical Similarity (ROUGE-L)	0.37 $\pm$ 0.06	-0.42 $\pm$ 0.16	<u>0.97$\pm$0.02</u>	0.67 $\pm$ 0.02	0.46 $\pm$ 0.08	0.27 $\pm$ 0.10	0.55 $\pm</$					

Table 4: AUROC $\uparrow$, PRR $\uparrow$, RCE $\downarrow$ for **Mistral-7B-Instruct** across tasks (fixed metric-wise gradient; warmer = higher value). Uncertainty estimators and AlignScore variants are ranked separately: bold = best per column (ties marked); underline = second-best (ties marked) within the uncertainty group; AlignScore group shows only the best.

Uncertainty Metric	RAC/Truth		RCE ↓		PreciseWiki		AURUC ↑		LongWiki		NonExistent	
	AURC ↑	PRR ↓	AURC ↑	PRR ↓	AURC ↑	PRR ↓	AURC ↑	PRR ↓	AURC ↑	PRR ↓	AURC ↑	PRR ↓
Informations	Maximum Seq. Probability (MSP)	0.76±0.01	0.62±0.03	0.09±0.05	0.76±0.02	0.61±0.05	0.19±0.07	0.52±0.02	0.13±0.07	0.61±0.12	0.68±0.03	0.44±0.09
	Perplexity (PPL)	0.56±0.02	0.01±0.04	0.49±0.06	0.71±0.02	0.61±0.05	0.17±0.06	0.53±0.02	0.14±0.07	0.58±0.12	0.55±0.03	0.15±0.10
	Mean Token Entropy (MTE)	0.53±0.02	-0.07±0.04	0.60±0.06	0.71±0.02	0.61±0.05	0.17±0.06	0.52±0.02	0.08±0.07	0.61±0.12	0.56±0.03	0.17±0.10
	Pairwise Mutual Info. (PMI)	0.35±0.01	-0.37±0.04	0.09±0.03	0.47±0.02	0.09±0.03	0.40±0.02	0.49±0.02	0.07±0.01	0.48±0.02	0.44±0.03	-0.24±0.13
	Conditional PMI	0.42±0.01	-0.37±0.04	0.95±0.06	0.28±0.02	-0.83±0.07	1.00±0.00	0.47±0.02	-0.08±0.09	0.85±0.06	0.44±0.03	0.23±0.11
	Self Conditionality	0.49±0.02	-0.18±0.04	0.69±0.05	0.57±0.02	0.30±0.06	0.42±0.10	0.52±0.02	0.05±0.08	0.62±0.14	0.60±0.03	0.24±0.10
	Rényi Divergence	0.46±0.02	-0.24±0.05	0.78±0.06	0.55±0.02	0.23±0.06	0.49±0.09	0.52±0.02	0.04±0.08	0.65±0.13	0.59±0.03	0.23±0.11
	Fisher-Rao Distance	0.50±0.02	-0.15±0.04	0.65±0.07	0.55±0.02	0.23±0.06	0.44±0.11	0.52±0.02	0.06±0.08	0.65±0.13	0.61±0.03	0.24±0.10
	Tokens SAR	0.56±0.02	0.00±0.04	0.50±0.06	0.75±0.02	0.66±0.05	0.13±0.06	0.53±0.02	0.14±0.07	0.57±0.12	0.55±0.03	0.13±0.11
	Claim-Conditioned Probability	0.76±0.01	0.57±0.03	0.09±0.05	0.74±0.02	0.62±0.05	0.20±0.06	0.59±0.02	0.35±0.07	0.39±0.09	0.71±0.03	0.51±0.09
Sampling-based	Monte-Carlo Seq. Entropy	0.34±0.01	-0.56±0.04	1.00±0.01	0.73±0.02	0.58±0.05	0.19±0.06	0.52±0.02	0.01±0.08	0.66±0.13	0.66±0.03	0.44±0.09
	Monte-Carlo Norm. Seq. Entropy	0.34±0.01	-0.57±0.04	1.00±0.01	0.72±0.02	0.62±0.05	0.14±0.06	0.51±0.02	0.00±0.08	0.71±0.12	0.60±0.03	0.29±0.09
	Semantic Entropy	0.34±0.01	-0.52±0.04	1.00±0.01	0.74±0.02	0.63±0.05	0.26±0.07	0.53±0.02	0.07±0.08	0.68±0.13	0.66±0.03	0.43±0.09
	Semantic Density	0.39±0.01	-0.78±0.04	0.98±0.05	0.78±0.02	0.71±0.06	0.12±0.06	0.57±0.02	0.22±0.07	0.42±0.11	0.43±0.03	0.18±0.08
	Sentence SAR	0.40±0.02	-0.44±0.04	0.94±0.04	0.77±0.02	0.68±0.05	0.20±0.07	0.55±0.02	0.00±0.08	0.69±0.10	0.72±0.03	0.54±0.08
	SAR	0.35±0.01	-0.48±0.04	1.00±0.01	0.76±0.02	0.66±0.04	0.16±0.06	0.56±0.02	0.19±0.07	0.47±0.12	0.53±0.02	0.15±0.10
	Cocosa MSP	0.75±0.01	0.58±0.03	0.08±0.05	0.77±0.02	0.66±0.05	0.21±0.06	0.56±0.02	0.22±0.07	0.49±0.10	0.62±0.03	0.38±0.09
	Cocosa PTE	0.55±0.02	0.01±0.04	0.52±0.08	0.74±0.02	0.63±0.05	0.17±0.07	0.57±0.02	0.22±0.07	0.39±0.11	0.50±0.03	0.04±0.10
Cocosa MLE	0.54±0.02	-0.04±0.04	0.55±0.07	0.75±0.02	0.63±0.05	0.18±0.07	0.57±0.02	0.21±0.07	0.39±0.11	0.51±0.03	0.07±0.10	
Internal	Attention Score	0.73±0.01	0.51±0.04	0.23±0.05	0.63±0.02	0.35±0.06	0.31±0.08	0.48±0.02	-0.07±0.08	0.77±0.01	0.70±0.03	0.47±0.09
	RAUQ	0.56±0.02	0.02±0.04	0.48±0.06	0.76±0.02	0.64±0.05	0.16±0.07	0.52±0.02	0.14±0.07	0.61±0.12	0.56±0.03	0.18±0.10
	CISL	0.42±0.02	-0.39±0.04	0.90±0.05	0.69±0.02	0.52±0.05	0.21±0.07	0.49±0.02	-0.02±0.08	0.77±0.01	0.48±0.03	-0.05±0.11
	Eigenscore	0.32±0.01	-0.60±0.04	1.00±0.01	0.68±0.02	0.37±0.06	0.41±0.11	0.48±0.02	-0.16±0.09	0.80±0.11	0.66±0.03	0.39±0.10
Training-based	Mahalanobis D.	0.24±0.01	-0.68±0.04	1.00±0.01	0.53±0.02	0.60±0.07	0.66±0.04	0.45±0.02	-0.15±0.08	0.92±0.08	0.26±0.02	1.01±0.12
	Mahalanobis D. RDE	0.39±0.01	-0.05±0.04	0.94±0.04	0.02±0.09	-0.51±0.05	0.72±0.10	0.57±0.02	-0.01±0.08	0.95±0.07	0.27±0.02	0.94±0.10
	HUQ-MD	0.56±0.02	0.01±0.04	0.49±0.06	0.71±0.02	0.61±0.05	0.17±0.06	0.53±0.02	0.14±0.07	0.57±0.12	0.55±0.03	0.15±0.10
	HUQ-RMD	0.55±0.02	0.00±0.04	0.49±0.07	0.71±0.02	0.61±0.05	0.17±0.06	0.53±0.02	0.14±0.07	0.57±0.12	0.55±0.03	0.15±0.10
	Relative	P(Truth)	0.48±0.01	-0.19±0.04	0.73±0.06	0.48±0.02	0.04±0.07	0.79±0.09	0.48±0.02	-0.12±0.08	0.83±0.11	0.26±0.03
P(Truth) Sampling		0.37±0.01	-0.34±0.04	1.00±0.01	0.54±0.02	0.19±0.06	0.49±0.12	-	-	-	-	-
P(Truth) Empirical		0.51±0.00	0.01±0.04	0.69±0.15	0.51±0.01	0.01±0.07	0.69±0.16	0.50±0.00	0.00±0.08	0.75±0.15	0.49±0.00	-0.02±0.11
NumSet		0.46±0.01	-0.10±0.04	0.92±0.08	0.78±0.01	0.75±0.04	0.11±0.05	0.53±0.01	0.13±0.08	0.58±0.14	0.44±0.02	-0.29±0.11
KelProb		0.34±0.01	-0.48±0.04	1.00±0.01	0.76±0.02	0.64±0.05	0.17±0.06	0.52±0.02	0.05±0.08	0.63±0.14	0.59±0.02	0.21±0.10
Black-box	Kernel Language Entropy (KLE)	0.33±0.01	-0.50±0.04	0.79±0.02	0.74±0.04	0.11±0.06	0.56±0.02	0.27±0.07	0.39±0.11	0.50±0.03	-0.05±0.11	0.74±0.11
	EigVallLP NLI (Entail)	0.33±0.01	-0.50±0.04	1.00±0.01	0.80±0.01	0.76±0.04	0.15±0.06	0.56±0.02	0.22±0.07	0.41±0.11	0.53±0.03	0.13±0.10
	EigVallLP NLI (Contra)	0.39±0.01	-0.25±0.04	1.00±0.01	0.76±0.02	0.70±0.05	0.15±0.06	0.61±0.02	0.35±0.07	0.36±0.09	0.47±0.03	-0.10±0.10
	EigVallLP Jaccard	0.30±0.01	-0.66±0.04	1.00±0.01	0.68±0.02	0.54±0.05	0.07±0.07	0.55±0.02	0.18±0.07	0.58±0.11	0.47±0.03	-0.27±0.12
	Eccentricity NLI (Entail)	0.33±0.01	-0.77±0.04	1.00±0.01	0.77±0.02	0.75±0.04	0.15±0.06	0.55±0.02	0.11±0.07	0.57±0.12	0.53±0.03	0.13±0.10
	Eccentricity NLI (Contra)	0.40±0.01	-0.25±0.04	1.00±0.01	0.79±0.01	0.75±0.04	0.13±0.05	0.62±0.02	0.32±0.07	0.30±0.06	0.46±0.03	-0.17±0.10
	Eccentricity Jaccard	0.48±0.01	-0.04±0.04	0.84±0.12	0.72±0.02	0.56±0.05	0.21±0.07	0.50±0.02	0.02±0.08	0.71±0.15	0.51±0.02	0.04±0.11
	DegMat NLI (Entail)	0.33±0.01	-0.51±0.04	1.00±0.01	0.80±0.01	0.75±0.04	0.13±0.06	0.55±0.02	0.18±0.08	0.54±0.12	0.55±0.03	0.20±0.10
	DegMat NLI (Contra)	0.37±0.01	-0.33±0.04	1.00±0.01	0.76±0.02	0.71±0.04	0.12±0.05	0.61±0.02	0.31±0.07	0.32±0.09	0.46±0.03	-0.12±0.10
	DegMat Jaccard	0.30±0.01	-0.65±0.04	1.00±0.01	0.70±0.02	0.52±0.05	0.19±0.07	0.55±0.02	0.19±0.07	0.49±0.11	0.44±0.03	-0.25±0.12
AlgoScore	AS-Precision(gold_answer)	0.66±0.01	0.22±0.04	0.42±0.05	0.73±0.02	0.34±0.06	0.67±0.08	0.61±0.02	0.34±0.07	0.35±0.08	0.71±0.03	0.49±0.09
	AS-Precision(wikipedia)	-	-	-	0.85±0.01	0.85±0.03	0.13±0.06	0.60±0.02	0.40±0.06	0.36±0.08	-	-
	AS-Recall(gold_answer)	0.58±0.01	0.04±0.04	0.56±0.06	0.91±0.01	0.81±0.04	0.43±0.11	0.59±0.02	0.36±0.07	0.37±0.10	0.72±0.03	0.52±0.08
	AS-Recall(wikipedia)	-	-	-	0.49±0.02	-0.05±0.07	0.79±0.13	0.50±0.02	0.22±0.07	0.49±0.12	-	-

Table 5: AUROC $\uparrow$, PRR $\uparrow$, RCE $\downarrow$ for **Llama-2-7B-Chat** across tasks (fixed metric-wise gradient; warmer = higher value). Uncertainty estimators and AlignScore variants are ranked separately: bold = best per column (ties marked); underline = second-best (ties marked) within the uncertainty group; AlignScore group shows only the best.

Uncertainty Metric	AUROC $\uparrow$	RAGTruth PRR $\uparrow$	RCE $\downarrow$	PreciseWiki PRR $\uparrow$	RCE $\downarrow$	LongWiki PRR $\uparrow$	RCE $\downarrow$	NonExistent PRR $\uparrow$	RCE $\downarrow$				
Maximum Seq. Probability (MSP)	0.53 $\pm$ 0.01	-0.01 $\pm$ 0.03	0.65 $\pm$ 0.08	0.74 $\pm$ 0.02	0.62 $\pm$ 0.07	0.30 $\pm$ 0.07	0.56 $\pm$ 0.01	0.22 $\pm$ 0.04	0.42 $\pm$ 0.08	0.57 $\pm$ 0.04	0.13 $\pm$ 0.14	0.61 $\pm$ 0.12	
Perplexity (PPL)	0.41 $\pm$ 0.01	-0.28 $\pm$ 0.04	0.99$\pm$0.02	0.68 $\pm$ 0.02	0.66 $\pm$ 0.06	0.21 $\pm$ 0.07	0.59 $\pm$ 0.01	0.27 $\pm$ 0.04	0.24 $\pm$ 0.08	0.49 $\pm$ 0.04	-0.11 $\pm$ 0.16	0.73 $\pm$ 0.12	
Mean Token Entropy (MTE)	0.53 $\pm$ 0.01	-0.02 $\pm$ 0.04	0.59 $\pm$ 0.09	0.64 $\pm$ 0.02	0.61 $\pm$ 0.08	0.22 $\pm$ 0.09	0.59 $\pm$ 0.01	0.28 $\pm$ 0.04	0.30 $\pm$ 0.08	0.51 $\pm$ 0.04	-0.01 $\pm$ 0.16	0.72 $\pm$ 0.11	
Pointwise Mutual Info. (PMI)	0.32 $\pm$ 0.01	-0.46 $\pm$ 0.04	1.00$\pm$0.00	0.46 $\pm$ 0.03	-0.22 $\pm$ 0.11	0.88$\pm$0.10	0.53 $\pm$ 0.01	0.12 $\pm$ 0.05	0.51 $\pm$ 0.12	0.46 $\pm$ 0.03	-0.02 $\pm$ 0.14	0.78 $\pm$ 0.08	
Conditional PMI	0.55 $\pm$ 0.01	0.06 $\pm$ 0.03	0.53 $\pm$ 0.10	0.29 $\pm$ 0.02	-0.84 $\pm$ 0.11	1.00$\pm$0.01	0.41 $\pm$ 0.01	-0.23 $\pm$ 0.05	1.00$\pm$0.01	0.51 $\pm$ 0.04	-0.06 $\pm$ 0.15	0.70 $\pm$ 0.12	
Self Certainty	0.46 $\pm$ 0.01	-0.20 $\pm$ 0.04	0.87 $\pm$ 0.05	0.59 $\pm$ 0.02	0.52 $\pm$ 0.07	0.49 $\pm$ 0.08	0.57 $\pm$ 0.01	0.20 $\pm$ 0.04	0.29 $\pm$ 0.09	0.52 $\pm$ 0.04	-0.01 $\pm$ 0.16	0.64 $\pm$ 0.11	
Rényi Divergence	0.46 $\pm$ 0.01	-0.21 $\pm$ 0.04	0.93 $\pm$ 0.05	0.58 $\pm$ 0.02	0.44 $\pm$ 0.07	0.52 $\pm$ 0.08	0.56 $\pm$ 0.01	0.18 $\pm$ 0.04	0.32 $\pm$ 0.10	0.51 $\pm$ 0.04	-0.07 $\pm$ 0.16	0.59 $\pm$ 0.11	
Fisher-Rao Distance	0.46 $\pm$ 0.01	-0.21 $\pm$ 0.04	0.85 $\pm$ 0.05	0.58 $\pm$ 0.02	0.46 $\pm$ 0.07	0.54 $\pm$ 0.08	0.57 $\pm$ 0.01	0.23 $\pm$ 0.04	0.28 $\pm$ 0.09	0.54 $\pm$ 0.04	0.04 $\pm$ 0.16	0.57 $\pm$ 0.09	
Token SAR	0.41 $\pm$ 0.01	-0.28 $\pm$ 0.04	0.99$\pm$0.02	0.74 $\pm$ 0.02	0.74$\pm$0.05	<u>0.16$\pm$0.06</u>	0.59 $\pm$ 0.01	0.25 $\pm$ 0.04	0.23 $\pm$ 0.08	0.50 $\pm$ 0.04	-0.10 $\pm$ 0.16	0.69 $\pm$ 0.12	
Chain-Conditioned Probability	0.49 $\pm$ 0.01	-0.03 $\pm$ 0.03	0.80 $\pm$ 0.07	0.72 $\pm$ 0.02	0.61 $\pm$ 0.08	0.26 $\pm$ 0.07	0.63 $\pm$ 0.01	0.39 $\pm$ 0.04	0.22 $\pm$ 0.07	<u>0.64$\pm$0.04</u>	<u>0.36$\pm$0.13</u>	0.42 $\pm$ 0.11	
Monte-Carlo Seq. Entropy	0.46 $\pm$ 0.01	-0.26 $\pm$ 0.03	0.82 $\pm$ 0.06	0.71 $\pm$ 0.02	0.63 $\pm$ 0.07	0.19 $\pm$ 0.07	0.51 $\pm$ 0.01	0.03 $\pm$ 0.05	0.65 $\pm$ 0.11	0.56 $\pm$ 0.04	0.16 $\pm$ 0.14	0.65 $\pm$ 0.10	
Monte-Carlo Norm. Seq. Entropy	0.38 $\pm$ 0.01	-0.37 $\pm$ 0.03	1.00$\pm$0.00	0.70 $\pm$ 0.02	0.63 $\pm$ 0.07	0.21 $\pm$ 0.07	0.54 $\pm$ 0.01	0.06 $\pm$ 0.05	0.53 $\pm$ 0.11	0.44 $\pm$ 0.04	-0.29 $\pm$ 0.16	0.90$\pm$0.10	
Semantic Entropy	0.46 $\pm$ 0.01	-0.27 $\pm$ 0.03	0.84 $\pm$ 0.05	0.73 $\pm$ 0.02	0.63 $\pm$ 0.07	0.22 $\pm$ 0.08	0.51 $\pm$ 0.01	0.04 $\pm$ 0.05	0.65 $\pm$ 0.15	0.55 $\pm$ 0.04	-0.15 $\pm$ 0.14	0.63 $\pm$ 0.11	
Semantic Density	0.48 $\pm$ 0.01	-0.07 $\pm$ 0.04	0.84 $\pm$ 0.11	0.71 $\pm$ 0.03	0.45 $\pm$ 0.08	0.45 $\pm$ 0.11	0.58 $\pm$ 0.01	0.23 $\pm$ 0.08	0.28 $\pm$ 0.09	0.49 $\pm$ 0.03	-0.04 $\pm$ 0.14	0.79$\pm$0.03	
Sentence SAR	0.56 $\pm$ 0.01	-0.02 $\pm$ 0.04	0.57 $\pm$ 0.05	0.75 $\pm$ 0.02	0.64 $\pm$ 0.07	0.27 $\pm$ 0.08	0.49 $\pm$ 0.01	0.00 $\pm$ 0.05	0.79$\pm$0.12	0.64$\pm$0.03	0.52$\pm$0.10	0.82$\pm$0.10	
SAR	0.42 $\pm$ 0.01	-0.28 $\pm$ 0.04	0.97$\pm$0.04	0.75 $\pm$ 0.02	0.66 $\pm$ 0.07	0.27 $\pm$ 0.07	0.61 $\pm$ 0.02	0.22 $\pm$ 0.04	0.35 $\pm$ 0.08	0.39 $\pm$ 0.03	-0.42 $\pm$ 0.16	0.95 $\pm$ 0.06	
Cocoa MSP	0.50 $\pm$ 0.01	-0.01 $\pm$ 0.03	0.76 $\pm$ 0.10	0.76$\pm$0.02	0.66 $\pm$ 0.07	0.27 $\pm$ 0.08	0.63 $\pm$ 0.01	0.35 $\pm$ 0.04	0.18 $\pm$ 0.07	0.49 $\pm$ 0.04	-0.14 $\pm$ 0.16	0.78 $\pm$ 0.10	
Cocoa PPL	0.42 $\pm$ 0.01	-0.27 $\pm$ 0.04	0.99 $\pm$ 0.03	0.72 $\pm$ 0.02	0.70 $\pm$ 0.06	0.17 $\pm$ 0.07	0.65 $\pm$ 0.01	0.39 $\pm$ 0.04	0.16 $\pm$ 0.06	0.44 $\pm$ 0.04	-0.31 $\pm$ 0.16	0.87 $\pm$ 0.09	
Cocoa MTE	0.50 $\pm$ 0.01	-0.07 $\pm$ 0.04	0.72 $\pm$ 0.08	0.70 $\pm$ 0.02	0.71 $\pm$ 0.06	0.16$\pm$0.07	0.65 $\pm$ 0.01	0.39 $\pm$ 0.04	0.15$\pm$0.06	0.45 $\pm$ 0.04	-0.20 $\pm$ 0.16	0.86 $\pm$ 0.10	
Attention Score	0.65$\pm$0.01	0.30$\pm$0.03	0.30$\pm$0.06	0.56 $\pm$ 0.02	0.30 $\pm$ 0.10	0.55 $\pm$ 0.11	0.45 $\pm$ 0.01	-0.16 $\pm$ 0.05	0.95$\pm$0.06	0.60 $\pm$ 0.04	0.24 $\pm$ 0.13	0.50 $\pm$ 0.13	
RAUQ	0.42 $\pm$ 0.01	-0.27 $\pm$ 0.04	0.98 $\pm$ 0.03	0.73 $\pm$ 0.02	0.66 $\pm$ 0.06	0.20 $\pm$ 0.07	0.58 $\pm$ 0.01	0.25 $\pm$ 0.04	0.28 $\pm$ 0.08	0.50 $\pm$ 0.04	-0.10 $\pm$ 0.16	0.72 $\pm$ 0.10	
CSL	0.37 $\pm$ 0.01	-0.34 $\pm$ 0.03	1.00$\pm$0.00	0.59 $\pm$ 0.02	0.38 $\pm$ 0.09	0.39 $\pm$ 0.11	0.56 $\pm$ 0.02	0.18 $\pm$ 0.05	0.36 $\pm$ 0.10	0.45 $\pm$ 0.03	-0.07 $\pm$ 0.14	0.85$\pm$0.11	
EigenScore	0.48 $\pm$ 0.01	-0.17 $\pm$ 0.03	0.83 $\pm$ 0.07	0.64 $\pm$ 0.03	0.29 $\pm$ 0.10	0.48 $\pm$ 0.11	0.42 $\pm$ 0.01	-0.21 $\pm$ 0.05	0.99$\pm$0.02	0.59 $\pm$ 0.04	0.13 $\pm$ 0.15	0.62 $\pm$ 0.10	
Mahalanobis D.	0.33 $\pm$ 0.01	-0.31 $\pm$ 0.04	1.00$\pm$0.00	0.52 $\pm$ 0.03	0.05 $\pm$ 0.10	0.65 $\pm$ 0.15	0.40 $\pm$ 0.01	-0.28 $\pm$ 0.05	1.00$\pm$0.01	0.18 $\pm$ 0.03	-1.49 $\pm$ 0.13	1.00$\pm$0.01	
Relative Mahalanobis D.	0.33 $\pm$ 0.01	-0.32 $\pm$ 0.04	1.00$\pm$0.00	0.52 $\pm$ 0.03	0.04 $\pm$ 0.11	0.66 $\pm$ 0.15	0.42 $\pm$ 0.01	-0.23 $\pm$ 0.05	0.99$\pm$0.03	0.18 $\pm$ 0.03	-1.49 $\pm$ 0.14	1.00$\pm$0.01	
RDE	0.33 $\pm$ 0.01	-0.28 $\pm$ 0.04	0.99 $\pm$ 0.02	0.55 $\pm$ 0.03	0.21 $\pm$ 0.09	0.53 $\pm$ 0.12	0.44 $\pm$ 0.01	-0.20 $\pm$ 0.05	1.00$\pm$0.02	0.25 $\pm$ 0.02	-1.06 $\pm$ 0.12	1.00$\pm$0.00	
HUQ-MD	0.41 $\pm$ 0.01	-0.28 $\pm$ 0.04	0.99 $\pm$ 0.02	0.68 $\pm$ 0.02	0.66 $\pm$ 0.06	0.19 $\pm$ 0.07	0.59 $\pm$ 0.01	0.27 $\pm$ 0.04	0.24 $\pm$ 0.08	0.49 $\pm$ 0.04	-0.11 $\pm$ 0.16	0.73 $\pm$ 0.12	
HUQ-RMD	0.41 $\pm$ 0.01	-0.28 $\pm$ 0.04	0.99$\pm$0.02	0.68 $\pm$ 0.02	0.66 $\pm$ 0.06	0.19 $\pm$ 0.07	0.59 $\pm$ 0.01	0.27 $\pm$ 0.04	0.24 $\pm$ 0.08	0.49 $\pm$ 0.04	-0.11 $\pm$ 0.16	0.73 $\pm$ 0.12	
Reflexive	P(Tru)	0.43 $\pm$ 0.01	-0.19 $\pm$ 0.04	0.98$\pm$0.03	0.58 $\pm$ 0.03	0.31 $\pm$ 0.10	0.44 $\pm$ 0.12	0.49 $\pm$ 0.02	-0.06 $\pm$ 0.05	0.83$\pm$0.10	0.39 $\pm$ 0.04	-0.54 $\pm$ 0.17	0.93$\pm$0.07
P(Tru) Sampling	0.50 $\pm$ 0.01	-0.02 $\pm$ 0.04	0.72 $\pm$ 0.12	0.53 $\pm$ 0.03	0.09 $\pm$ 0.10	0.62 $\pm$ 0.13	-	-	-	-	-	-	
P(Tru) Empirical	0.52 $\pm$ 0.01	0.08 $\pm$ 0.03	0.53 $\pm$ 0.11	0.44 $\pm$ 0.02	-0.11 $\pm$ 0.10	0.87$\pm$0.08	0.49 $\pm$ 0.02	-0.08 $\pm$ 0.05	0.85$\pm$0.12	0.54 $\pm$ 0.03	0.10 $\pm$ 0.14	0.60 $\pm$ 0.15	
NumSet	0.48 $\pm$ 0.01	-0.08 $\pm$ 0.04	0.81 $\pm$ 0.12	0.73 $\pm$ 0.02	0.60 $\pm$ 0.07	0.22 $\pm$ 0.08	0.51 $\pm$ 0.01	0.05 $\pm$ 0.05	0.62 $\pm$ 0.14	0.45 $\pm$ 0.03	-0.23 $\pm$ 0.16	0.84 $\pm$ 0.12	
LabelProb	0.52 $\pm$ 0.01	0.01 $\pm$ 0.04	0.58 $\pm$ 0.11	0.71 $\pm$ 0.02	0.53 $\pm$ 0.08	0.28 $\pm$ 0.09	0.52 $\pm$ 0.01	0.06 $\pm$ 0.05	0.57 $\pm$ 0.14	0.49 $\pm$ 0.03	-0.05 $\pm$ 0.15	0.77 $\pm$ 0.13	
Kernel Language Entropy (KLE)	0.47 $\pm$ 0.01	-0.14 $\pm$ 0.04	0.85$\pm$0.07	0.70 $\pm$ 0.03	0.45 $\pm$ 0.09	0.41 $\pm$ 0.11	0.58 $\pm$ 0.02	0.20 $\pm$ 0.05	0.29 $\pm$ 0.08	0.50 $\pm$ 0.04	-0.08 $\pm$ 0.16	0.74 $\pm$ 0.10	
EigValLap NLI (Entail)	0.47 $\pm$ 0.01	-0.14 $\pm$ 0.04	0.83 $\pm$ 0.10	0.72 $\pm$ 0.03	0.52 $\pm$ 0.08	0.34 $\pm$ 0.09	0.57 $\pm$ 0.02	0.16 $\pm$ 0.05	0.38 $\pm$ 0.09	0.47 $\pm$ 0.04	-0.13 $\pm$ 0.16	0.80 $\pm$ 0.12	
EigValLap NLI (Contra)	0.43 $\pm$ 0.01	-0.09 $\pm$ 0.03	0.95 $\pm$ 0.05	0.60 $\pm$ 0.03	0.22 $\pm$ 0.09	0.55 $\pm$ 0.11	0.65$\pm$0.01	0.40$\pm$0.04	0.15$\pm$0.06	0.53 $\pm$ 0.04	0.11 $\pm$ 0.15	0.66 $\pm$ 0.13	
EigValLap Jaccard	0.37 $\pm$ 0.01	-0.38 $\pm$ 0.04	1.00$\pm$0.00	0.65 $\pm$ 0.03	0.40 $\pm$ 0.09	0.38 $\pm$ 0.09	0.60 $\pm$ 0.01	0.25 $\pm$ 0.04	0.28 $\pm$ 0.08	0.32 $\pm$ 0.03	-0.70 $\pm$ 0.15	1.00$\pm$0.01	
Eccentricity NLI (Entail)	0.51 $\pm$ 0.01	-0.02 $\pm$ 0.04	0.68 $\pm$ 0.12	0.72 $\pm$ 0.03	0.53 $\pm$ 0.09	0.33 $\pm$ 0.09	0.57 $\pm$ 0.02	0.14 $\pm$ 0.05	0.41 $\pm$ 0.10	0.49 $\pm$ 0.03	0.04 $\pm$ 0.15	0.75 $\pm$ 0.11	
Eccentricity NLI (Contra)	0.43 $\pm$ 0.01	-0.11 $\pm$ 0.04	0.96$\pm$0.05	0.70 $\pm$ 0.02	0.54 $\pm$ 0.08	0.30 $\pm$ 0.09	0.64 $\pm$ 0.01	0.36 $\pm$ 0.04	0.17$\pm$0.07	0.54 $\pm$ 0.04	0.17 $\pm$ 0.15	0.61 $\pm$ 0.12	
Eccentricity Jaccard	0.56$\pm$0.01	0.09 $\pm$ 0.03	0.46 $\pm$ 0.11	0.67 $\pm$ 0.03	0.37 $\pm$ 0.09	0.40 $\pm$ 0.09	0.48 $\pm$ 0.01	-0.06 $\pm$ 0.05	0.89$\pm$0.12	0.52 $\pm$ 0.03	0.08 $\pm$ 0.14	0.67 $\pm$ 0.15	
DegMat NLI (Entail)	0.48 $\pm$ 0.01	-0.12 $\pm$ 0.04	0.78 $\pm$ 0.09	0.73 $\pm$ 0.03	0.54 $\pm$ 0.08	0.37 $\pm$ 0.10	0.55 $\pm$ 0.02	0.12 $\pm$ 0.05	0.45 $\pm$ 0.11	0.48 $\pm$ 0.04	-0.11 $\pm$ 0.16	0.80 $\pm$ 0.13	
DegMat NLI (Contra)	0.41 $\pm$ 0.01	-0.15 $\pm$ 0.04	0.99$\pm$0.08	0.62 $\pm$ 0.03	0.26 $\pm$ 0.10	0.49 $\pm$ 0.12	0.65 $\pm$ 0.01	0.38 $\pm$ 0.04	0.15$\pm$0.06	0.53 $\pm$ 0.04	0.05 $\pm$ 0.15	0.65 $\pm$ 0.15	
DegMat Jaccard	0.40 $\pm$ 0.01	-0.34 $\pm$ 0.04	0.97 $\pm$ 0.03	0.67 $\pm$ 0.03	0.43 $\pm$ 0.09	0.39 $\pm$ 0.09	0.61 $\pm$ 0.01	0.28 $\pm$ 0.04	0.24 $\pm$ 0.08	0.33 $\pm$ 0.03	-0.69 $\pm$ 0.15	1.00$\pm$0.02	
LUQ	0.37 $\pm$ 0.01	-0.30 $\pm$ 0.04	1.00$\pm$0.01	0.71 $\pm$ 0.03	0.49 $\pm$ 0.08	0.38 $\pm$ 0.11	0.65 $\pm$ 0.01	0.35 $\pm$ 0.04	0.21 $\pm$ 0.08	0.49 $\pm$ 0.04	-0.16 $\pm$ 0.16	0.75 $\pm$ 0.11	
Lexical Sim. (ROUGE-L)	0.40 $\pm$ 0.01	-0.34 $\pm$ 0.04	0.96 $\pm$ 0.04	0.64 $\pm$ 0.03	0.38 $\pm$ 0.10	0.40 $\pm$ 0.11	0.56 $\pm$ 0.01	0.17 $\pm$ 0.05	0.35 $\pm$ 0.10	0.25 $\pm$ 0.03	-1.10 $\pm$ 0.14	1.00$\pm$0.00	
Lexical Sim. (BLEU)	0.42 $\pm$ 0.01	-0.31 $\pm$ 0.04	0.95 $\pm$ 0.04	0.66 $\pm$ 0.03	0.45 $\pm$ 0.09	0.31 $\pm$ 0.09	0.60 $\pm$ 0.01	0.26 $\pm$ 0.04	0.21 $\pm$ 0.07	0.25 $\pm$ 0.03	-1.17 $\pm$ 0.14	1.00$\pm$0.01	
AlignScore	AS-Precision(gold_answer)	0.71$\pm$0.01	0.42$\pm$0.03	0.11$\pm$0.05	0.88 $\pm$ 0.02	0.82 $\pm$ 0.06	0						

Table 6: AUROC $\uparrow$, PRR $\uparrow$, RCE $\downarrow$ for **Llama-2-13B-Chat** across tasks (fixed metric-wise gradient; warmer = higher value). Uncertainty estimators and AlignScore variants are ranked separately: bold = best per column (ties marked); underline = second-best (ties marked) within the uncertainty group; AlignScore group shows only the best.

Uncertainty Metric	RAGTruth			PreciseWiki			LongWiki			NonExistent			
	AUROC $\uparrow$	PRR $\uparrow$	RCE $\downarrow$	AUROC $\uparrow$	PRR $\uparrow$	RCE $\downarrow$	AUROC $\uparrow$	PRR $\uparrow$	RCE $\downarrow$	AUROC $\uparrow$	PRR $\uparrow$	RCE $\downarrow$	
Information-based	Maximum Seq. Probability (MSP)	0.63 $\pm$ 0.01	0.23 $\pm$ 0.03	0.24 $\pm$ 0.05	0.64 $\pm$ 0.02	0.44 $\pm$ 0.06	0.34 $\pm$ 0.07	0.59 $\pm$ 0.02	0.28 $\pm$ 0.06	0.30 $\pm$ 0.09	0.76 $\pm$ 0.02	0.58 $\pm$ 0.05	0.18 $\pm$ 0.07
	Perplexity (PPL)	0.35 $\pm$ 0.01	-0.40 $\pm$ 0.03	1.00 $\pm$ 0.00	0.55 $\pm$ 0.02	0.09 $\pm$ 0.07	0.60 $\pm$ 0.09	0.59 $\pm$ 0.02	0.28 $\pm$ 0.06	0.29 $\pm$ 0.09	0.25 $\pm$ 0.02	-1.00 $\pm$ 0.07	1.00 $\pm$ 0.00
	Mean Token Entropy (MTE)	0.51 $\pm$ 0.01	-0.15 $\pm$ 0.03	0.75 $\pm$ 0.04	0.56 $\pm$ 0.02	0.14 $\pm$ 0.07	0.57 $\pm$ 0.09	0.59 $\pm$ 0.02	0.27 $\pm$ 0.06	0.29 $\pm$ 0.09	0.26 $\pm$ 0.02	-0.93 $\pm$ 0.07	1.00 $\pm$ 0.00
	Pointwise Mutual Info. (PMI)	0.31 $\pm$ 0.01	-0.46 $\pm$ 0.03	1.00 $\pm$ 0.01	0.47 $\pm$ 0.02	-0.18 $\pm$ 0.08	0.87 $\pm$ 0.09	0.49 $\pm$ 0.02	-0.04 $\pm$ 0.07	0.76 $\pm$ 0.12	0.42 $\pm$ 0.02	-0.07 $\pm$ 0.07	0.85 $\pm$ 0.05
	Conditional PMI	0.56 $\pm$ 0.01	0.04 $\pm$ 0.03	0.50 $\pm$ 0.07	0.34 $\pm$ 0.02	-0.54 $\pm$ 0.08	0.96 $\pm$ 0.04	0.40 $\pm$ 0.02	-0.35 $\pm$ 0.07	0.98 $\pm$ 0.04	0.75 $\pm$ 0.02	0.60 $\pm$ 0.05	0.19 $\pm$ 0.06
	Self Certainty	0.43 $\pm$ 0.01	-0.30 $\pm$ 0.03	0.91 $\pm$ 0.05	0.54 $\pm$ 0.02	0.20 $\pm$ 0.07	0.54 $\pm$ 0.09	0.55 $\pm$ 0.02	0.13 $\pm$ 0.07	0.50 $\pm$ 0.11	0.40 $\pm$ 0.02	-0.45 $\pm$ 0.07	0.75 $\pm$ 0.11
	Rényi Divergence	0.42 $\pm$ 0.01	-0.31 $\pm$ 0.03	0.95 $\pm$ 0.04	0.53 $\pm$ 0.02	0.17 $\pm$ 0.07	0.62 $\pm$ 0.11	0.54 $\pm$ 0.02	0.10 $\pm$ 0.07	0.51 $\pm$ 0.12	0.41 $\pm$ 0.02	-0.42 $\pm$ 0.07	0.72 $\pm$ 0.09
	Fisher-Rao Distance	0.44 $\pm$ 0.01	-0.29 $\pm$ 0.03	0.91 $\pm$ 0.04	0.51 $\pm$ 0.02	0.10 $\pm$ 0.07	0.66 $\pm$ 0.09	0.56 $\pm$ 0.02	0.16 $\pm$ 0.07	0.47 $\pm$ 0.10	0.45 $\pm$ 0.02	-0.34 $\pm$ 0.07	0.59 $\pm$ 0.07
	Token SAR	0.35 $\pm$ 0.01	-0.40 $\pm$ 0.03	1.00 $\pm$ 0.00	0.63 $\pm$ 0.02	0.30 $\pm$ 0.07	0.36 $\pm$ 0.10	0.60 $\pm$ 0.02	0.31 $\pm$ 0.06	0.29 $\pm$ 0.09	0.21 $\pm$ 0.02	-1.19 $\pm$ 0.07	1.00 $\pm$ 0.01
	Claim-Conditioned Probability	0.58 $\pm$ 0.01	0.13 $\pm$ 0.03	0.31 $\pm$ 0.10	0.70 $\pm$ 0.02	0.61 $\pm$ 0.05	0.17 $\pm$ 0.06	0.64 $\pm$ 0.02	0.46 $\pm$ 0.06	0.23 $\pm$ 0.07	0.85 $\pm$ 0.01	0.79 $\pm$ 0.04	0.16 $\pm$ 0.07
Sampling-based	Monte-Carlo Seq. Entropy	0.46 $\pm$ 0.01	-0.26 $\pm$ 0.03	0.80 $\pm$ 0.05	0.58 $\pm$ 0.02	0.22 $\pm$ 0.07	0.42 $\pm$ 0.12	0.54 $\pm$ 0.02	0.12 $\pm$ 0.07	0.52 $\pm$ 0.12	0.52 $\pm$ 0.02	0.10 $\pm$ 0.06	0.58 $\pm$ 0.13
	Monte-Carlo Norm. Seq. Entropy	0.38 $\pm$ 0.01	-0.33 $\pm$ 0.03	0.99 $\pm$ 0.02	0.58 $\pm$ 0.02	0.24 $\pm$ 0.07	0.47 $\pm$ 0.10	0.55 $\pm$ 0.02	0.13 $\pm$ 0.07	0.51 $\pm$ 0.12	0.40 $\pm$ 0.02	-0.28 $\pm$ 0.06	0.99 $\pm$ 0.02
	Semantic Entropy	0.49 $\pm$ 0.01	-0.16 $\pm$ 0.04	0.77 $\pm$ 0.04	0.59 $\pm$ 0.02	0.27 $\pm$ 0.07	0.39 $\pm$ 0.11	0.58 $\pm$ 0.02	0.24 $\pm$ 0.06	0.40 $\pm$ 0.10	0.51 $\pm$ 0.02	0.06 $\pm$ 0.06	0.68 $\pm$ 0.13
	Semantic Density	0.49 $\pm$ 0.01	-0.05 $\pm$ 0.03	0.78 $\pm$ 0.11	0.75 $\pm$ 0.02	0.64 $\pm$ 0.03	0.14 $\pm$ 0.06	0.62 $\pm$ 0.02	0.42 $\pm$ 0.06	0.27 $\pm$ 0.08	0.33 $\pm$ 0.02	-0.58 $\pm$ 0.06	1.00 $\pm$ 0.00
	Sentence SAR	0.58 $\pm$ 0.01	0.04 $\pm$ 0.03	0.53 $\pm$ 0.04	0.63 $\pm$ 0.02	0.45 $\pm$ 0.06	0.34 $\pm$ 0.06	0.54 $\pm$ 0.02	0.14 $\pm$ 0.07	0.53 $\pm$ 0.12	0.61 $\pm$ 0.02	0.34 $\pm$ 0.05	0.28 $\pm$ 0.08
	SAR	0.46 $\pm$ 0.01	-0.19 $\pm$ 0.03	0.82 $\pm$ 0.06	0.65 $\pm$ 0.02	0.41 $\pm$ 0.06	0.28 $\pm$ 0.08	0.61 $\pm$ 0.02	0.29 $\pm$ 0.06	0.33 $\pm$ 0.10	0.36 $\pm$ 0.02	-0.42 $\pm$ 0.07	1.00 $\pm$ 0.01
	Cocoa MSP	0.60 $\pm$ 0.01	0.15 $\pm$ 0.03	0.34 $\pm$ 0.09	0.69 $\pm$ 0.02	0.58 $\pm$ 0.05	0.22 $\pm$ 0.06	0.64 $\pm$ 0.02	0.43 $\pm$ 0.06	0.21 $\pm$ 0.07	0.68 $\pm$ 0.02	0.42 $\pm$ 0.05	0.27 $\pm$ 0.09
	Cocoa PPL	0.38 $\pm$ 0.01	-0.37 $\pm$ 0.03	1.00 $\pm$ 0.00	0.60 $\pm$ 0.02	0.31 $\pm$ 0.06	0.43 $\pm$ 0.08	0.65 $\pm$ 0.02	0.44 $\pm$ 0.06	0.21 $\pm$ 0.07	0.23 $\pm$ 0.02	-1.09 $\pm$ 0.06	1.00 $\pm$ 0.00
	Cocoa MTE	0.48 $\pm$ 0.01	-0.19 $\pm$ 0.03	0.78 $\pm$ 0.05	0.61 $\pm$ 0.02	0.36 $\pm$ 0.06	0.35 $\pm$ 0.09	0.64 $\pm$ 0.02	0.42 $\pm$ 0.06	0.18 $\pm$ 0.07	0.24 $\pm$ 0.02	-1.02 $\pm$ 0.07	1.00 $\pm$ 0.00
	Attention Score	0.69 $\pm$ 0.01	0.41 $\pm$ 0.03	0.30 $\pm$ 0.05	0.58 $\pm$ 0.02	0.29 $\pm$ 0.06	0.39 $\pm$ 0.10	0.47 $\pm$ 0.02	-0.13 $\pm$ 0.07	0.90 $\pm$ 0.10	0.87 $\pm$ 0.01	0.74 $\pm$ 0.04	0.35 $\pm$ 0.07
Inter-rater	RAUQ	0.35 $\pm$ 0.01	-0.40 $\pm$ 0.03	1.00 $\pm$ 0.00	0.56 $\pm$ 0.02	0.14 $\pm$ 0.07	0.67 $\pm$ 0.09	0.58 $\pm$ 0.02	0.29 $\pm$ 0.06	0.32 $\pm$ 0.10	0.27 $\pm$ 0.02	-0.93 $\pm$ 0.07	1.00 $\pm$ 0.00
	CSL	0.38 $\pm$ 0.01	-0.38 $\pm$ 0.03	0.99 $\pm$ 0.01	0.60 $\pm$ 0.02	0.29 $\pm$ 0.06	0.44 $\pm$ 0.08	0.54 $\pm$ 0.02	0.13 $\pm$ 0.07	0.50 $\pm$ 0.12	0.49 $\pm$ 0.02	-0.35 $\pm$ 0.07	0.79 $\pm$ 0.08
	EigenScore	0.51 $\pm$ 0.01	-0.10 $\pm$ 0.03	0.78 $\pm$ 0.04	0.53 $\pm$ 0.02	0.03 $\pm$ 0.07	0.62 $\pm$ 0.11	0.50 $\pm$ 0.02	0.08 $\pm$ 0.07	0.72 $\pm$ 0.12	0.42 $\pm$ 0.02	-0.20 $\pm$ 0.06	0.98 $\pm$ 0.04
		0.23 $\pm$ 0.01	-0.44 $\pm$ 0.03	1.00 $\pm$ 0.00	0.47 $\pm$ 0.02	-0.01 $\pm$ 0.07	0.78 $\pm$ 0.10	0.35 $\pm$ 0.02	-0.53 $\pm$ 0.08	1.00 $\pm$ 0.00	0.60 $\pm$ 0.02	0.19 $\pm$ 0.07	0.45 $\pm$ 0.07
Training-based	Relative Mahalanobis D.	0.25 $\pm$ 0.01	-0.49 $\pm$ 0.03	1.00 $\pm$ 0.00	0.46 $\pm$ 0.02	-0.05 $\pm$ 0.07	0.84 $\pm$ 0.10	0.36 $\pm$ 0.02	-0.49 $\pm$ 0.08	1.00 $\pm$ 0.00	0.52 $\pm$ 0.02	0.06 $\pm$ 0.06	0.65 $\pm$ 0.10
	RDE	0.25 $\pm$ 0.01	-0.37 $\pm$ 0.03	1.00 $\pm$ 0.01	0.59 $\pm$ 0.02	0.37 $\pm$ 0.06	0.33 $\pm$ 0.09	0.46 $\pm$ 0.02	-0.12 $\pm$ 0.07	0.92 $\pm$ 0.08	0.52 $\pm$ 0.02	0.18 $\pm$ 0.05	0.69 $\pm$ 0.05
	HUQ-MD	0.35 $\pm$ 0.01	-0.40 $\pm$ 0.03	1.00 $\pm$ 0.00	0.55 $\pm$ 0.02	0.07 $\pm$ 0.07	0.62 $\pm$ 0.09	0.59 $\pm$ 0.02	0.28 $\pm$ 0.06	0.29 $\pm$ 0.09	0.25 $\pm$ 0.02	-1.00 $\pm$ 0.07	1.00 $\pm$ 0.00
	HUQ-RMD	0.35 $\pm$ 0.01	-0.40 $\pm$ 0.03	1.00 $\pm$ 0.00	0.55 $\pm$ 0.02	0.07 $\pm$ 0.07	0.62 $\pm$ 0.09	0.59 $\pm$ 0.02	0.28 $\pm$ 0.06	0.29 $\pm$ 0.09	0.25 $\pm$ 0.02	-1.00 $\pm$ 0.07	1.00 $\pm$ 0.00
		0.63 $\pm$ 0.01	0.25 $\pm$ 0.03	0.19 $\pm$ 0.07	0.46 $\pm$ 0.02	-0.05 $\pm$ 0.07	0.87 $\pm$ 0.10	0.45 $\pm$ 0.02	-0.15 $\pm$ 0.07	0.95 $\pm$ 0.07	0.61 $\pm$ 0.02	0.44 $\pm$ 0.06	0.52 $\pm$ 0.05
	P(True) Sampling	0.63 $\pm$ 0.01	0.26 $\pm$ 0.03	0.19 $\pm$ 0.07	0.46 $\pm$ 0.02	-0.16 $\pm$ 0.08	0.78 $\pm$ 0.10	-	-	-	-	-	-
Reflective	P(True) Empirical	0.67 $\pm$ 0.01	0.40 $\pm$ 0.03	0.19 $\pm$ 0.06	0.52 $\pm$ 0.02	0.04 $\pm$ 0.07	0.61 $\pm$ 0.15	0.49 $\pm$ 0.02	-0.05 $\pm$ 0.07	0.80 $\pm$ 0.14	0.33 $\pm$ 0.02	-0.60 $\pm$ 0.06	1.00 $\pm$ 0.00
	NumSet	0.47 $\pm$ 0.01	-0.09 $\pm$ 0.03	0.81 $\pm$ 0.13	0.70 $\pm$ 0.02	0.57 $\pm$ 0.05	0.13 $\pm$ 0.06	0.55 $\pm$ 0.01	0.21 $\pm$ 0.06	0.55 $\pm$ 0.13	0.45 $\pm$ 0.01	-0.21 $\pm$ 0.06	0.93 $\pm$ 0.08
	LabelProb	0.51 $\pm$ 0.01	0.00 $\pm$ 0.03	0.62 $\pm$ 0.11	0.66 $\pm$ 0.02	0.40 $\pm$ 0.06	0.28 $\pm$ 0.08	0.59 $\pm$ 0.02	0.27 $\pm$ 0.06	0.32 $\pm$ 0.10	0.45 $\pm$ 0.02	-0.14 $\pm$ 0.06	0.96 $\pm$ 0.06
Black-box	Kernel Language Entropy (KLE)	0.49 $\pm$ 0.01	-0.04 $\pm$ 0.03	0.71 $\pm$ 0.11	0.72 $\pm$ 0.02	0.62 $\pm$ 0.05	0.18 $\pm$ 0.07	0.64 $\pm$ 0.02	0.42 $\pm$ 0.06	0.21 $\pm$ 0.07	0.40 $\pm$ 0.02	-0.28 $\pm$ 0.06	0.98 $\pm$ 0.03
	EigValLap NLI (Entail)	0.51 $\pm$ 0.01	0.03 $\pm$ 0.03	0.66 $\pm$ 0.13	0.71 $\pm$ 0.02	0.64 $\pm$ 0.05	0.15 $\pm$ 0.06	0.62 $\pm$ 0.02	0.35 $\pm$ 0.06	0.23 $\pm$ 0.08	0.40 $\pm$ 0.02	-0.28 $\pm$ 0.06	0.99 $\pm$ 0.02
	EigValLap NLI (Contra)	0.43 $\pm$ 0.01	-0.16 $\pm$ 0.03	0.98 $\pm$ 0.04	0.66 $\pm$ 0.02	0.41 $\pm$ 0.06	0.25 $\pm$ 0.09	0.68 $\pm$ 0.02	0.50 $\pm$ 0.06	0.17 $\pm$ 0.07	0.44 $\pm$ 0.02	-0.16 $\pm$ 0.06	0.97 $\pm$ 0.05
	EigValLap Jaccard	0.42 $\pm$ 0.01	-0.27 $\pm$ 0.03	0.95 $\pm$ 0.04	0.63 $\pm$ 0.02	0.35 $\pm$ 0.06	0.30 $\pm$ 0.09	0.60 $\pm$ 0.02	0.31 $\pm$ 0.06	0.30 $\pm$ 0.09	0.40 $\pm$ 0.02	-0.32 $\pm$ 0.06	1.00 $\pm$ 0.01
	Eccentricity NLI (Entail)	0.52 $\pm$ 0.01	0.04 $\pm$ 0.03	0.59 $\pm$ 0.10	0.67 $\pm$ 0.02	0.40 $\pm$ 0.07	0.31 $\pm$ 0.08	0.62 $\pm$ 0.02	0.34 $\pm$ 0.06	0.29 $\pm$ 0.08	0.44 $\pm$ 0.02	-0.13 $\pm$ 0.06	0.93 $\pm$ 0.07
	Eccentricity NLI (Contra)	0.42 $\pm$ 0.01	-0.17 $\pm$ 0.03	0.99 $\pm$ 0.03	0.71 $\pm$ 0.02	0.58 $\pm$ 0.05	0.18 $\pm$ 0.07	0.67 $\pm$ 0.02	0.44 $\pm$ 0.06	0.23 $\pm$ 0.07	0.44 $\pm$ 0.02	-0.17 $\pm$ 0.06	0.97 $\pm$ 0.05
	Eccentricity Jaccard	0.57 $\pm$ 0.01	0.09 $\pm$ 0.03	0.57 $\pm$ 0.11	0.61 $\pm$ 0.02	0.29 $\pm$ 0.07	0.29 $\pm$ 0.10	0.49 $\pm$ 0.02	-0.03 $\pm$ 0.07	0.78 $\pm$ 0.15	0.53 $\pm$ 0.01	0.11 $\pm$ 0.06	0.53 $\pm$ 0.14
	DegMat NLI (Entail)	0.51 $\pm$ 0.01	0.02 $\pm$ 0.03	0.63 $\pm$ 0.13	0.73 $\pm$ 0.02	0.64 $\pm$ 0.05	0.13 $\pm$ 0.06	0.62 $\pm$ 0.02	0.34 $\pm$ 0.06	0.24 $\pm$ 0.08	0.40 $\pm$ 0.02	-0.28 $\pm$ 0.06	0.99 $\pm$ 0.02
	DegMat NLI (Contra)	0.41 $\pm$ 0.01	-0.20 $\pm$ 0.03	0.99 $\pm$ 0.02	0.67 $\pm$ 0.02	0.46 $\pm$ 0.06	0.30 $\pm$ 0.07	0.68 $\pm$ 0.02	0.51 $\pm$ 0.05	0.19 $\pm$ 0.07	0.44 $\pm$ 0.02	-0.19 $\pm$ 0.06	0.96 $\pm$ 0.06
	DegMat Jaccard	0.45 $\pm$ 0.01	-0.21 $\pm$ 0.03	0.85 $\pm$ 0.05	0.63 $\pm$ 0.02	0.36 $\pm$ 0.06	0.24 $\pm$ 0.09	0.60 $\pm$ 0.02	0.32 $\pm$ 0.06	0.29 $\pm$ 0.09	0.40 $\pm$ 0.02	-0.34 $\pm$ 0.07	1.00 $\pm$ 0.01
	LUUQ	0.43 $\pm$ 0.01	-0.15 $\pm$ 0.03	0.93 $\pm$ 0.06	0.73 $\pm$ 0.02	0.63 $\pm$ 0.05	0.17 $\pm$ 0.07	0.68 $\pm$ 0.02	0.51 $\pm$ 0.05	0.21 $\pm$ 0.07	0.40 $\pm$ 0.02	-0.32 $\pm$ 0.06	0.98 $\pm$ 0.04
	Lexical Sim. (ROUGE-L)	0.43 $\pm$ 0.01	-0.29 $\pm$ 0.03	0.94 $\pm$ 0.04	0.65 $\pm$ 0.02	0.43 $\pm$ 0.06	0.21 $\pm$ 0.08	0.55 $\pm$ 0.02	0.16 $\pm$ 0.06	0.45 $\pm$ 0.11	0.34 $\pm$ 0.02	-0.53 $\pm$ 0.07	1.00 $\pm$ 0.00
Lexical Sim. (BLEU)	0.46 $\pm$ 0.01	-0.21 $\pm$ 0.03	0.86 $\pm$ 0.06	0.61 $\pm$ 0.02	0.32 $\pm$ 0.06	0.32 $\pm$ 0.10	0.58 $\pm$ 0.02	0.25 $\pm$ 0.06	0.36 $\pm$ 0.10	0.32 $\pm$ 0.02	-0.62 $\pm$ 0.06	1.00 $\pm$ 0.00	
Align-Score	AS-Precision(gold_answer)	0.74 $\pm$ 0.01	0.47 $\pm$										

Table 7: Mean estimator ranks across benchmark panels, pooling all tasks and models. Within each panel, estimators are ranked by the corresponding metric, with rank 1 indicating the best score. Each cell reports the mean rank, with the subscript indicating one standard deviation across panels; lower is better. Parentheses indicate the number of panels in which the estimator ranks in the top 3. Bold marks the best mean rank within uncertainty estimators and within AlignScore variants separately; underline marks the second-best uncertainty estimator.

Estimator	AUROC	PRR	RCE
Maximum Seq. Probability (MSP)	13.3 $\pm$ 10.3 (2)	13.8 $\pm$ 9.0 (1)	14.3 $\pm$ 10.0 (2)
Perplexity (PPL)	27.4 $\pm$ 9.9 (0)	26.2 $\pm$ 11.3 (0)	22.7 $\pm$ 12.7 (0)
Mean Token Entropy (MTE)	23.7 $\pm$ 9.8 (0)	23.1 $\pm$ 9.1 (0)	20.4 $\pm$ 9.3 (0)
Pointwise Mutual Info. (PMI)	40.1 $\pm$ 7.4 (0)	39.8 $\pm$ 10.7 (0)	36.8 $\pm$ 9.3 (0)
Conditional PMI	32.6 $\pm$ 18.1 (0)	32.7 $\pm$ 17.7 (1)	32.5 $\pm$ 18.8 (2)
Self Certainty	28.1 $\pm$ 8.9 (0)	28.5 $\pm$ 7.8 (0)	24.0 $\pm$ 10.8 (0)
Rényi Divergence	30.4 $\pm$ 9.0 (0)	30.8 $\pm$ 7.3 (0)	26.6 $\pm$ 12.7 (0)
Fisher–Rao Distance	26.8 $\pm$ 11.0 (0)	27.6 $\pm$ 8.4 (0)	25.6 $\pm$ 11.7 (0)
Token SAR	24.1 $\pm$ 11.7 (0)	24.2 $\pm$ 12.1 (0)	20.6 $\pm$ 13.2 (1)
Claim-Conditioned Probability (CCP)	8.9 $\pm$ 6.1 (4)	7.8 $\pm$ 6.4 (5)	10.3 $\pm$ 7.5 (4)
Monte-Carlo Seq. Entropy	24.8 $\pm$ 10.7 (0)	26.2 $\pm$ 12.4 (0)	20.7 $\pm$ 12.6 (1)
Monte-Carlo Norm. Seq. Entropy	31.8 $\pm$ 7.8 (0)	31.6 $\pm$ 10.0 (0)	30.8 $\pm$ 12.5 (0)
Semantic Entropy	22.1 $\pm$ 9.6 (0)	24.5 $\pm$ 10.5 (0)	19.9 $\pm$ 9.8 (1)
Semantic Density	20.2 $\pm$ 10.9 (0)	20.7 $\pm$ 10.6 (0)	22.9 $\pm$ 12.4 (1)
Sentence SAR	17.2 $\pm$ 14.5 (2)	17.8 $\pm$ 13.6 (2)	18.5 $\pm$ 13.5 (1)
SAR	22.2 $\pm$ 9.9 (0)	24.8 $\pm$ 8.4 (0)	25.8 $\pm$ 9.7 (0)
Cocoa MSP	9.9 $\pm$ 5.1 (1)	11.3 $\pm$ 7.5 (1)	14.3 $\pm$ 9.6 (1)
Cocoa PPL	23.3 $\pm$ 13.2 (0)	21.9 $\pm$ 14.0 (0)	23.0 $\pm$ 13.3 (0)
Cocoa MTE	20.6 $\pm$ 11.2 (1)	19.8 $\pm$ 11.4 (0)	17.0 $\pm$ 12.5 (2)
Attention Score	21.9 $\pm$ 19.3 (3)	21.2 $\pm$ 18.5 (4)	22.2 $\pm$ 18.4 (2)
RAUQ	24.8 $\pm$ 10.7 (0)	23.9 $\pm$ 10.7 (0)	24.3 $\pm$ 13.4 (1)
CSL	32.8 $\pm$ 7.5 (0)	33.2 $\pm$ 5.7 (0)	31.6 $\pm$ 8.3 (0)
EigenScore	29.5 $\pm$ 14.3 (0)	31.3 $\pm$ 13.6 (0)	31.0 $\pm$ 12.0 (0)
Mahalanobis Distance (MD)	43.8 $\pm$ 10.4 (0)	42.7 $\pm$ 10.4 (0)	40.6 $\pm$ 9.9 (0)
Relative MD (RMD)	44.0 $\pm$ 9.1 (0)	42.1 $\pm$ 9.6 (0)	40.5 $\pm$ 9.4 (0)
Robust Density Estimation (RDE)	36.8 $\pm$ 13.2 (0)	33.2 $\pm$ 14.3 (0)	33.6 $\pm$ 13.1 (0)
HUQ-MD	28.0 $\pm$ 9.9 (0)	26.9 $\pm$ 11.6 (0)	22.7 $\pm$ 13.1 (0)
HUQ-RMD	28.3 $\pm$ 9.3 (0)	26.9 $\pm$ 11.1 (0)	22.9 $\pm$ 12.8 (0)
P(True)	34.8 $\pm$ 15.2 (0)	33.5 $\pm$ 14.9 (0)	33.0 $\pm$ 14.8 (1)
P(True) Sampling	30.5 $\pm$ 15.9 (0)	28.8 $\pm$ 17.4 (0)	29.2 $\pm$ 17.3 (1)
P(True) Empirical	30.4 $\pm$ 15.4 (1)	32.0 $\pm$ 17.1 (1)	30.9 $\pm$ 15.2 (1)
NumSet	23.8 $\pm$ 10.2 (0)	23.5 $\pm$ 10.6 (0)	26.7 $\pm$ 10.7 (0)
LabelProb	22.2 $\pm$ 8.9 (0)	22.8 $\pm$ 7.0 (0)	28.8 $\pm$ 8.3 (0)
Kernel Language Entropy (KLE)	19.8 $\pm$ 9.5 (0)	20.2 $\pm$ 9.3 (0)	20.8 $\pm$ 11.5 (1)
EigValLap NLI (entail)	20.3 $\pm$ 10.6 (1)	20.1 $\pm$ 10.3 (1)	20.9 $\pm$ 10.8 (2)
EigValLap NLI (contra)	18.4 $\pm$ 11.4 (2)	17.2 $\pm$ 11.3 (1)	23.8 $\pm$ 13.4 (0)
EigValLap Jaccard	31.2 $\pm$ 10.1 (0)	31.7 $\pm$ 10.2 (0)	29.7 $\pm$ 10.8 (0)
Eccentricity NLI (entail)	18.8 $\pm$ 7.8 (0)	20.7 $\pm$ 8.4 (0)	23.5 $\pm$ 8.2 (0)
Eccentricity NLI (contra)	17.2 $\pm$ 11.3 (1)	15.3 $\pm$ 9.1 (0)	21.8 $\pm$ 14.3 (1)
Eccentricity Jaccard	24.2 $\pm$ 12.4 (0)	24.7 $\pm$ 14.2 (1)	25.8 $\pm$ 13.3 (0)
DegMat NLI (entail)	20.2 $\pm$ 11.7 (0)	20.3 $\pm$ 10.3 (0)	19.2 $\pm$ 12.4 (2)
DegMat NLI (contra)	20.8 $\pm$ 12.6 (1)	18.5 $\pm$ 11.2 (1)	22.8 $\pm$ 16.2 (2)
DegMat Jaccard	29.4 $\pm$ 9.6 (0)	30.2 $\pm$ 10.7 (0)	26.6 $\pm$ 10.4 (1)
LUQ	20.5 $\pm$ 13.7 (0)	20.6 $\pm$ 12.7 (0)	21.6 $\pm$ 13.5 (1)
Lexical Sim. (ROUGE-L)	33.7 $\pm$ 8.1 (0)	34.5 $\pm$ 8.6 (0)	29.8 $\pm$ 9.3 (0)
Lexical Sim. (BLEU)	31.5 $\pm$ 9.0 (0)	33.6 $\pm$ 8.3 (0)	25.8 $\pm$ 11.7 (1)
AS-Precision(gold_answer)	5.3 $\pm$ 6.1 (6)	6.6 $\pm$ 9.8 (6)	16.5 $\pm$ 14.0 (1)
AS-Precision(wikipege)	2.7 $\pm$ 1.7 (5)	1.0 (6)	9.4 $\pm$ 4.6 (1)
AS-Recall(gold_answer)	7.4 $\pm$ 7.4 (6)	8.9 $\pm$ 8.5 (5)	20.9 $\pm$ 11.6 (0)
AS-Recall(wikipege)	34.2 $\pm$ 10.3 (0)	34.0 $\pm$ 11.6 (0)	33.5 $\pm$ 9.8 (0)

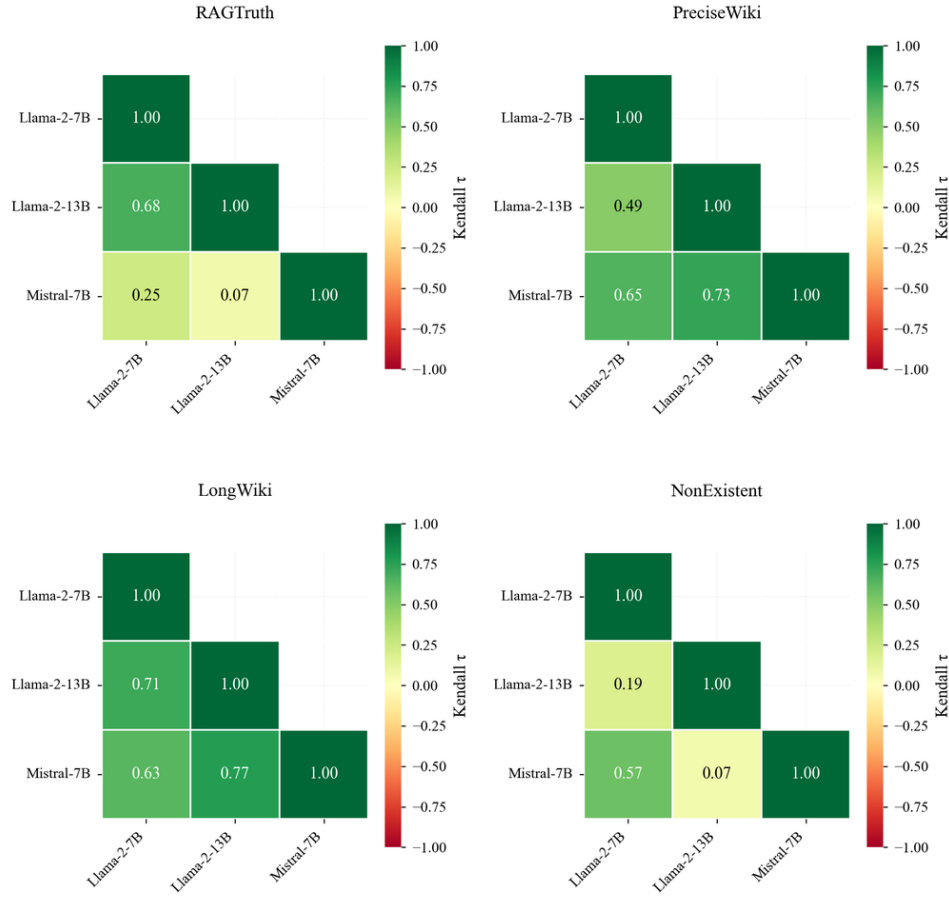


Figure 5: Pairwise Kendall's τ agreement between estimator rankings induced by ROC-AUC across models, shown separately for each task. Higher values indicate that the same estimators tend to rank similarly across models for a given hallucination type.

Table 8: Within-panel Spearman rank correlations between evaluation metrics, computed over uncertainty estimators. AlignScore variants are excluded.

Task	Model	AUROC-PRR	AUROC-RCE	PRR-RCE
RagTruth	Llama2-13b-chat	.96	-.99	-.95
–	Llama2-7b-chat	.94	-.97	-.91
–	Mistral-7B-instruct	.95	-.94	-.86
PreciseWikiQA	Llama2-13b-chat	.97	-.96	-.95
–	Llama2-7b-chat	.82	-.79	-.94
–	Mistral-7B-instruct	.98	-.81	-.88
LongWiki	Llama2-13b-chat	.99	-.98	-.97
–	Llama2-7b-chat	.98	-.96	-.97
–	Mistral-7B-instruct	.97	-.99	-.97
NonExistentRefusal	Llama2-13b-chat	.94	-.94	-.88
–	Llama2-7b-chat	.94	-.97	-.90
–	Mistral-7B-instruct-v0.2	.99	-.97	-.96
Median	–	.97	-.96	-.95
Range	–	[.82,.99]	[-.99, -.79]	[-.97, -.86]

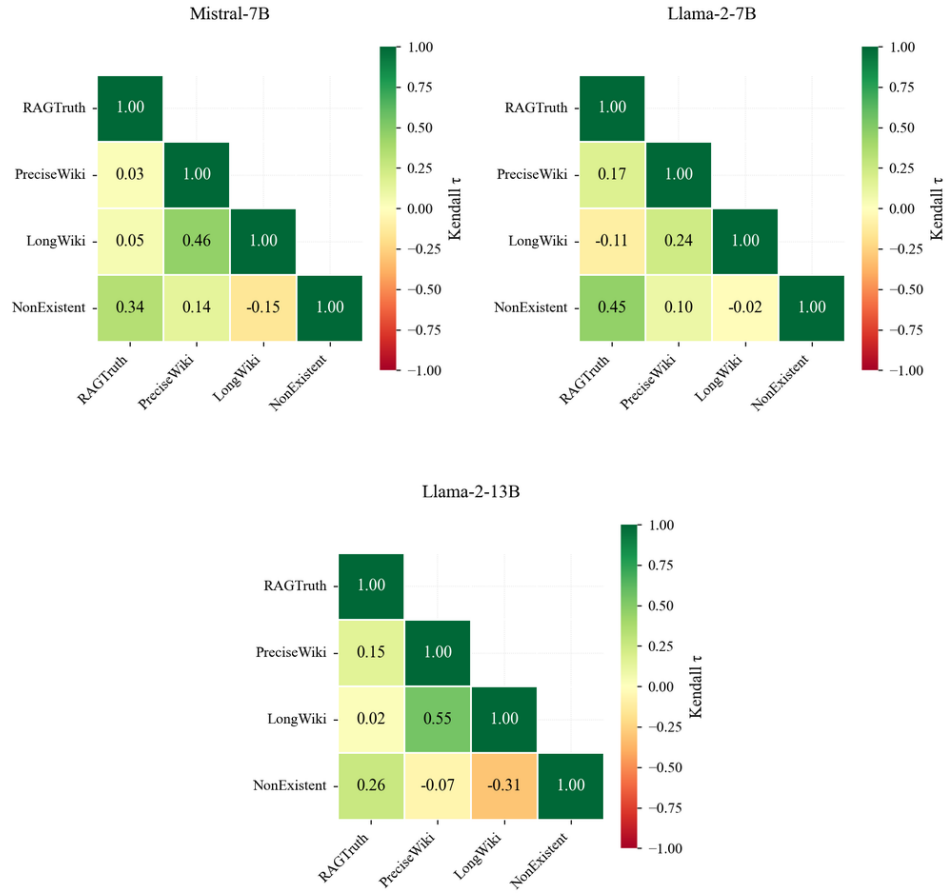


Figure 6: Pairwise Kendall’s τ agreement between estimator rankings induced by ROC-AUC across tasks, shown separately for each model. Higher values indicate that the same estimators tend to rank similarly across hallucination types for a given model.

Table 9: Dataset quality summary across models and tasks. H=Human annotation, J=Judge LLM.

Model	RAGTruth % Hallu. (H) ↓	PreciseWiki % Hallu. (J) ↓	LongWiki F1 @32 (J) ↑	NonExistent % Non-refusal (J) ↓
Llama 2 7B Chat	61.6	93.5	0.38 ± 0.19	96.4
Llama 2 13B Chat	56.7	87.5	0.31 ± 0.16	83.2
Mistral 7B Instruct	66.8	86.6	0.29 ± 0.16	93.8

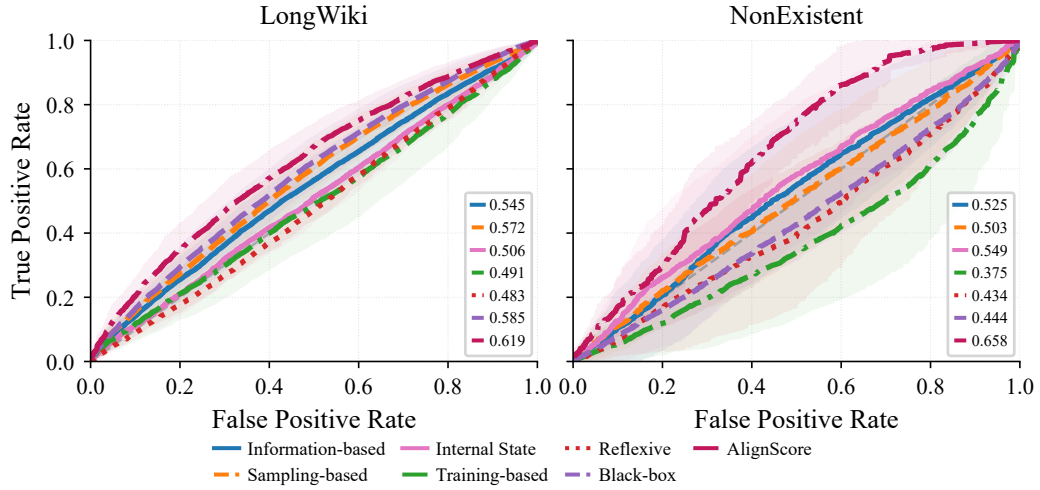


Figure 7: Family-level ROC aggregates per task, averaged across the three models. For each task–model panel, estimator ROC curves are first averaged within each family by taking the mean true positive rate at each false positive rate. For each task, these model-level family means are then averaged across models. Shaded bands show one standard deviation of these model-level family-mean true positive rates at each false positive rate; they reflect cross-model variability.